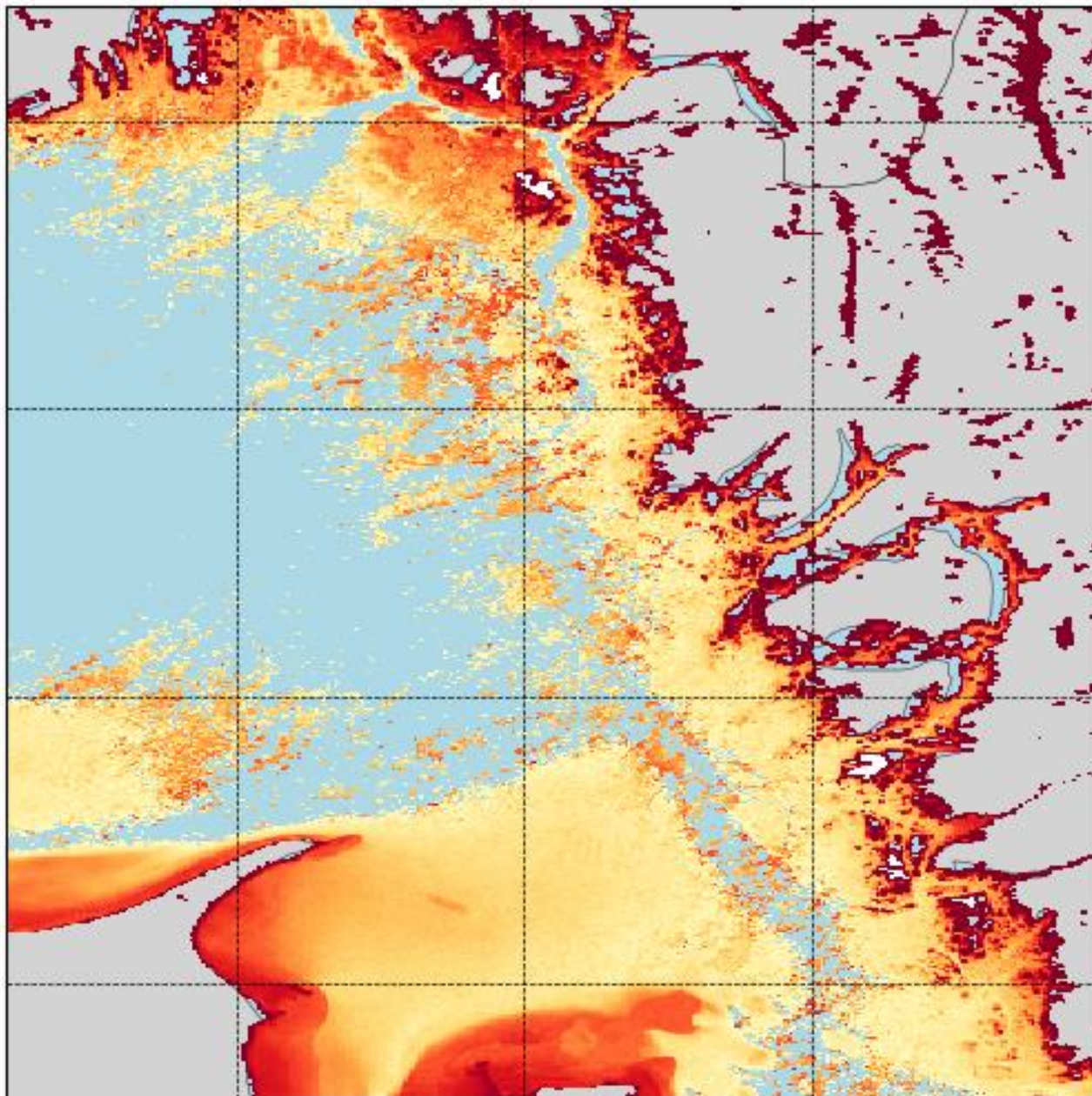


DIGITAL TWIN OCEAN – LYSEKILS MUNICIPALITY PILOT PROJECT

Technical Report



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Project information

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PROJECT SUMMARY

The overarching goal of the pilot project “Marine Planning – The Case of Lysekils Kommun” was to contribute to an open data framework for the monitoring of marine water using publicly available remotely sensed data. Marine planning has become more complex as human activity in coastal waters has increased while at the same time climate change shifts the underlying environmental conditions. This project aimed to assist in more flexible and better-informed marine planning efforts by contributing insights derived from continuously collected remote sensing data as well as expert experience. Undertaken in collaboration with Lysekil Kommun and Region Fyrbodal on the west coast of Sweden, the project has shown the feasibility of using remotely sensed data – specifically data from the Copernicus Earth observation system – to create near real-time data products that add dimensionality to the local and regional understanding of the coastal and offshore marine environments. In the future, these data products can improve marine planning efforts by providing local, regional, and national actors with up-to-date information on how specific environmental conditions are changing and how human activity is impacting the marine environment.

The project focused on three data products that were primarily built with remote sensing data accessed from the Copernicus Marine Data Store:

1. An investigation of the underwater light environment.
2. A machine learning model of macroalgae habitat distribution.
3. Water quality indicators including chlorophyll a concentration, oxygen availability at the seafloor, and water transparency.

All three data products were successfully generated for a spatial and temporal pilot study area in Sweden’s west coast marine area using the Copernicus WEkEO remote environment. The data products and methodologies are published in a series of Jupyter Notebooks in a GitHub repository ([link](#)) that serve as both tutorials and runnable code for non-expert users wishing to build the same data products for their own area of interest in the Baltic Sea drainage basin and time series.

The work done in this project shows the potential for remotely sensed data to be leveraged in a digital twin context to monitor and model key environmental and human factors in the marine environment. Based on the project team’s experience, it is advised that future work to achieve this goal focus on the following three areas:

1. Building the data and software infrastructure required to continuously produce and make data products such as those created in this project easily available. In essence, a functioning Digital Twin Ocean.
2. Refining and validating the scientific approach and algorithms used to derive data products through improved research. Many of the algorithms currently used vary by region but can be improved in the future using remotely sensed data.
3. Developing further data and modelling products in collaboration with stakeholders that can be included in a Digital Twin Ocean.

Table of Contents

1	INTRODUCTION	6
2	METHODOLOGY	6
2.1	TOOLS AND PLATFORMS	6
2.1.1.	WEkEO JupyterLabs Workspace	6
2.1.2	Copernicus Marine Services Data Store	7
2.1.3	Open-Source Python Libraries	7
2.2	DATA SOURCES	7
2.3	DATA PROCESSING AND ANALYTICAL METHODS	9
2.4	VALIDATION METHODS	10
3	DATA PRODUCT DESCRIPTION	10
3.1	EUPHOTIC ZONE DEPTH AND EUPHOTIC ZONE DELINEATION	10
3.2	MACROALGAE HABITAT DISTRIBUTION	11
3.2.1	Independent Variables	11
3.2.2	Dependent Variable	12
3.2.3	Model Training	12
3.3	WATER QUALITY INDICATORS	13
3.3.1	Chlorophyll a	13
3.3.2	Hypoxic Conditions	14
3.3.3	Secchi Depth	15
4	RESULTS	16
4.1	EUPHOTIC ZONE DEPTH AND EUPHOTIC ZONE DELINEATION	16
4.2	MACROALGAE HABITAT DISTRIBUTION	19
4.3	WATER QUALITY INDICATORS	23
4.3.1	Chlorophyll a	23
4.3.2	Hypoxic Conditions	27
4.3.3	Secchi Depth	29
5	DISCUSSION	36
	STRENGTHS AND LIMITATIONS OF THE APPROACH	36
	IMPLICATIONS FOR MARINE PLANNING, MONITORING AND POLICY	36
6	FURTHER DEVELOPMENT	37
	Deployment of a full-scale Digital Twin Ocean for Maritime Planning	37
	Scientific data product improvement	37



Further product development	37
Interactive maritime planning tools	38
7 REFERENCES	39
APPENDIX A: GITHUB REPOSITORY	41

1 INTRODUCTION

The pilot project “Havsplanering – Fall Lysekils Kommun” explores the development of methodologies and data products for a Digital Twin Ocean (DTO) in Sweden’s west coast marine area, including Skagerrak, Kattegatt, and the Danish straits, which form part of the transition zone and the drainage basin of the Baltic Sea. The concept of a DTO refers to a dynamic, data-driven digital representation of the marine environment that can support marine spatial planning, environmental monitoring, and decision-making. This pilot focuses on the application of open-source satellite data, particularly from the Copernicus Earth observation program, to generate actionable insights for local and regional stakeholders. The project is carried out in collaboration with the Swedish Space Agency and municipalities in northern Bohuslän. Funding comes from the Framework Partnership Agreement on Copernicus User Uptake (FPCUP).

The primary objective of the pilot is to demonstrate how open-access remote sensing data can be transformed into living, sensor-derived data products that are relevant for maritime planning. Specifically, the project has developed and tested three data products:

- Euphotic Zone Depth and Delineation
- Macroalgae Habitat Distribution
- Water Quality Indicators

These data products are intended to be integrated into a Digital Twin Ocean prototype, with the goal of delivering an open-source, scalable solution that can be adapted to other coastal regions.

As a pilot study, the scope of the project was intentionally focused on a subset of variables that were both feasible and aligned well with the project’s thematic goals. The developed data products were chosen in collaboration with municipal partners based on their practical applicability, data availability, and potential to effectively demonstrate the value of a DTO approach in regional planning context.

2 METHODOLOGY

The usage and analysis of remotely sensed data poses significant challenges to computing systems due to the size of datasets. This is especially relevant in the context of digital twins where the goal is to access time series and continuously updated data. This pilot has focused on investigating the feasibility of the creation of certain data products relevant to maritime planning and has thus been conducted in a remote environment using entirely open-source tools. The following section describes the tools and platforms used, data sources, data processing, and validation methods.

2.1 TOOLS AND PLATFORMS

The following tools and platforms were used in the pilot project – all data and tools are freely available to public users.

2.1.1. WEKEO JUPYTERLABS WORKSPACE

The project was completed entirely in a free WEkEO JupyterLabs Workspace – a suite of service for Copernicus data analysis and visualisation. WEkEO is the EU’s reference service for environmental data and provides a single access point for all Copernicus data on a collaborative platform.¹ With a free account, a WEkEO user can create and maintain a custom python environment that is deployed on the preexisting Copernicus organisation’s server infrastructure. The user can leverage this infrastructure to develop and maintain multiple Jupyter Notebooks and data analyses without downloading and manipulating large time

¹ Copernicus Programme, “WEkEO Platform.”

series datasets on their local IT infrastructure, allowing for lightweight research and proof of concept projects. In the DTO Skagerrak project a suite of Jupyter Notebooks was developed for specific data analysis products. These analyses are easily replicable by a non-experienced user using the annotated Jupyter Notebooks delivered with this report.

2.1.2 COPERNICUS MARINE SERVICES DATA STORE

The primary source of remote sensing data and their derived products used in this project were retrieved from the Copernicus Marine Service Data Store.² This service provides a wealth of living data products – including level 3 satellite data, multi-sensor aggregated level 4 satellite data, numerical models of the subsurface ocean environment, and in-situ data. With free login credentials created in the Copernicus Marine Services website, these datasets can be accessed via the CMEMS python API. Automating data loading via the API allows the user to subset datasets both spatially and temporally, as well as allowing for the use of dask arrays, greatly reducing the in-memory load of large dataset.

2.1.3 OPEN-SOURCE PYTHON LIBRARIES

This project primarily used the pre-built *wekeolab* conda environment provided in the WEkEO JupyterLab Workspace, adding a few external open-source python libraries to a custom conda environment for data access and manipulation. The full list of libraries used in this project can be found in the *DTO_Skagerrak_environment.txt* file found in the GitHub repository linked to in Appendix A. The environment file can be used to setup a custom conda environment without the need to reinstall external python packages. In brief, the project used *xcube-cmems* for data import and subsetting, *xarray*, *rioxarray*, and *bottleneck* for data cube investigation and manipulation, *geopandas* and *shapely* for geospatial analysis, and *cartopy*, *matplotlib*, and *xcube-viewer* for visualisation.

2.2 DATA SOURCES

The following data products from Copernicus Marine were used in the project, and can continue to be accessed in near real-time through the Copernicus Marine Data Store python API:

² Copernicus Programme, “Copernicus Marine Data Store.”

Table 1. CMEMS data products used for input data in the pilot project.

Dataset	CMEMS Product ID	Variables	Platform
Baltic Sea Ocean Colour Plankton, Reflectances, Transparency and Optics L3 NRT daily observations ³	OCEANCOLOUR_BAL_BGC_L3_NRT_009_131	• Kd490: Diffuse attenuation coefficient of light at 490 nm • RRS490: Surface ratio of upwelling radiance emerging from sea water to downwelling radiative flux in air at 490 nm • RRS708_75: Surface ratio of upwelling radiance emerging from sea water to downwelling radiative flux in air at 708.75 nm	OLCI-S3A & S3B
Baltic Sea Multiyear Ocean Colour Plankton, Reflectances and Transparency L3 daily observations ⁴	OCEANCOLOUR_BAL_BGC_L3_MY_009_133	• CHL: Mass concentration of chlorophyll a in sea water	OLCI-S3A & S3B
Baltic Sea- Sea Surface Temperature Reprocessed ⁵	DMI_BAL_SST_L4_REP_OBSERVATIONS_010_016	• analysed_sst: Analysed sea surface temperature	ESA CCI & Copernicus C3S
Baltic Sea Biogeochemistry Analysis and Forecast ⁶	BALTICSEA_ANALYSIS_FORECAST_BGC_003_007	• o2b: Mole concentration of dissolved molecular oxygen in sea water at seafloor	ERGOM & NEMO (numerical model)

³ Colella, S. et al., "Copernicus Marine Product User Manual for Ocean Colour Products"

⁴ Colella, S. et al., "Copernicus Marine Product User Manual for Ocean Colour Products"

⁵ Karagali, I. et al., "Product User Manual for Baltic L4 Sea- Sea Surface Temperature Reprocessed and Baltic Sea - L3S Sea Surface Temperature Reprocessed Products"

⁶ Lindenthal, A. et al., "Product User Manual for Physical Analysis and Forecasting Product"

In addition to the remote sensing data used in this project, four external open-source datasets were also used to create the analysis products. These datasets are included in the code repository, and are listed here:

- **Bathymetric data from the European Marine Observation and Data Network (EMODnet).** This is the best freely available full coverage bathymetric data available, with a spatial resolution of 1/16 arc minute. A dataset clipped to the project's area of interest is included in the code repository.⁷
- **Wave exposure from the Swedish Environmental Protection Agency.** Wave exposure is a measure of the general degree of wave action at a location. It has a local impact and is generally considered to be one of the most important factors influencing the composition of flora and fauna on shallow bottoms and shorelines.⁸
- **Water bodies – Coastal Waters from the Swedish Water Agency.** Coastal water zones are geographic areas along the Swedish coast used for the purposes of ecological health classification. Each zone has similar ecological conditions and thus unique reference values for varying conditions (such as chlorophyll a concentration, water transparency, etc.). These zones have been used in this project for the purposes of water quality indicator classification.⁹
- **Training data from SMHI SHARK database.** Swedish marine environmental monitoring data are stored by the Swedish Meteorological and Hydrological Institute in the database SHARK (Svenskt HavsARKiv). Epibenthic monitoring data from SHARK from diving and underwater video surveys were used as training data in the macroalgae habitat distribution model.¹⁰

2.3 DATA PROCESSING AND ANALYTICAL METHODS

This project used exclusively the WEkEO JupyterLabs Workspace for all data loading, subsetting, analysis, and result generation. The Jupyter Notebooks included in the GitHub repository linked in Appendix A can be loaded directly into a WEkEO Workspace and run by the user to generate their own results given the selected spatial and temporal subsetting parameters.

The general technical workflow of the Jupyter Notebooks data products is outlined below (note that the actual workflow varies somewhat depending on the requirements of the specific data product):

- 1) Requisite python libraries are installed in the WEkEO conda environment and imported in the notebook. All python libraries including version numbers can be found in the *DTO_Skagerrak_environment.txt* file found in the GitHub repository.
- 2) User login credentials are passed as variables in order to access the CMEMS data store.
- 3) Spatial and temporal dataset subsets are loaded into an xarray dataset using dask when required. Dask arrays allow for the reduction of a dataset's memory footprint by dividing the array into smaller pieces (chunks) that can fit into memory and be streamed as needed from the disk.
- 4) The raw data product from CMEMS is visualised in time and space.
- 5) Data manipulation on the xarray dataset is carried out. This step depends on the purpose of the data product and includes the application of algorithms from the literature to derive new parameters and variables from one or multiple CMEMS data store products, as well as machine learning algorithms in the case of modelling data products.
- 6) Required external data is loaded into the notebook after first being uploaded by the user to their WEkEO Workspace.

⁷ EMODnet Bathymetry consortium. "Download service for EMODnet Bathymetry Product tiles."

⁸ Albertsson, J., et al. "Sammanställning och Analys av Kustnära Undervattenmiljö (SAKU)."

⁹ Vattenmyndigheterna, "VM Vattenförekomster kustvatten 2016-2021."

¹⁰ SMHI, "SHARK."

- 7) Additional analytics are conducted to combine or filter the xarray datasets derived from the CMEMS data store with the external data sources.
- 8) Results are visualised in static plots as well as interactive maps to provide useful insights into the marine environment.

2.4 VALIDATION METHODS

This proof-of-concept project did not include ground truthing campaigns in its scope, and thus the data products are mainly derivative – generated using algorithms from the research literature in the Baltic Sea drainage basin. This research and the inherent assumptions are presented further in the data product descriptions and discussion sections of this report. The project team is aware of the limitations of data products that have not been verified with observational data, but it was deemed to be a more effective use of resources to build the data pipelines and show the feasibility of creating usable, easily accessible data products than conducting costly ground-truthing campaigns.

The exception to this is the machine learning model of macroalgae habitat zones, which is a validated modelled product as opposed to a derivative product. The macroalgae habitat model was built using a large observational dataset of epibenthic species cover and abundance spanning 2016-2024 from the database SHARK. During model training, 20% of the dataset was reserved for validation purposes. This data product is discussed in more detail in the data product description and results.

3 DATA PRODUCT DESCRIPTION

3.1 EUPHOTIC ZONE DEPTH AND EUPHOTIC ZONE DELINEATION

In this data product, the euphotic zone depth, Z_{eu} , - the portion of the water column where sunlight penetrates sufficiently to support photosynthesis – was modelled by integrating remote sensing optical depth data with high-resolution bathymetry. The euphotic zone is defined as the upper layer of the ocean where sunlight reaches a depth at which at least 1% of the surface irradiance remains.¹¹ This zone is considered ecologically significant, as it identifies the habitat for both phytoplankton in the water column and benthic vegetation on the seafloor. These photosynthetic organisms form the foundation of marine food webs and play a key role in global carbon cycling.¹² In addition, the depth of the euphotic zone may be essential for predicting algal bloom dynamics, particularly in nutrient rich waters.¹³

For this analysis, remote sensing data from the Copernicus Marine Service (CMEMS) were used, specifically the spectral diffuse attenuation coefficient at 490 nm, $K_d(490)^{-1}$ derived from Sentinel-3 OLCI observations (see Section 2.2). This variable quantifies how rapidly light diminishes with depth, particularly in the blue spectrum, which penetrates deepest in ocean water.¹⁴ Since light attenuation and depth are inversely related, this coefficient can be used to estimate the euphotic zone depth. For this project, an empirical formula adapted for the Baltic Sea is applied:¹⁵

$$Z_{eu} = 6.8 \cdot K_d(490)^{-1}$$

In this data product, the euphotic zone depth is defined, following Ryther (1956), as the vertical distance from the water surface to the depth where only 1% of surface light intensity remains. This definition is used to

¹¹ Ryther, "Photosynthesis in the Ocean as a Function of Light Intensity," 61-70

¹² Tett, P. "The photic zone." 59-87

¹³ Kratzer, S. et al., "Assessing Secchi and photic zone depth in the Baltic Sea from satellite data"

¹⁴ Austin, R. W., & Petzold, T. J., "The determination of the diffuse attenuation coefficient of sea water using the Coastal Zone Color Scanner."

¹⁵ Kratzer, S. et al., "Assessing Secchi and photic zone depth in the Baltic Sea from satellite data"

derive the variable `Zeu_depth`, which represents the modelled depth of the euphotic layer independent of water depth. The euphotic zone variable `Zeu_zone` is represented as a binary parameter and is defined as the seafloor areas where at least 1% of surface light reaches the bottom.

To delineate the euphotic zone (`Zeu_zone`), the calculated euphotic zone depth (`Zeu_depth`) is overlaid on EMODnet bathymetric data, allowing identification of area where the seafloor is within the euphotic zone. This dual approach enables both quantitative analysis of light penetration and spatial mapping of photosynthetically active areas, supporting assessment of phytoplankton productivity in the water column and benthic habitat availability on the seafloor. Additionally, `Zeu_percent`, a variable indicating the percentage of surface light reaching the seafloor at each pixel, was computed.

The model is designed to support flexible temporal analysis, enabling users to access and average data by day, week, or season. This facilitates the assessment of spatial and temporal variability in underwater light environments, which is essential for understanding habitat availability and primary productivity in marine ecosystems.

3.2 MACROALGAE HABITAT DISTRIBUTION

This data product presents a machine learning approach to modelling the distribution of kelp habitat zones. Large perennial macroalgae such as kelp are important ecosystem components in shallow coastal ecosystems where they often form dense belts on hard substrates and provide a complex habitat for a large number of species. Increased nutrition load and a following decrease in Secchi depth in coastal areas is a threat that will limit the distribution of macroalgae communities to a shallower depth, decreasing the total area covered by these communities. The spatial distribution and cover of macroalgae communities is therefore subject to change over time and depends on environmental factors that can be monitored using RS data. The distribution of kelp habitat, including the large habitat forming kelp species *Laminaria digitata*, *Laminaria hyperborea* and *Saccharina latissima* was modelled using random forest classification.

Kelp distribution is limited by environmental conditions such as light availability, seabed substrate, salinity and wave exposure. For spatial modelling and prediction of these species, raster layers of independent variables which cover the study area and have a spatial resolution which is fine enough for the spatial variation in heterogenic coastal environments are needed.

3.2.1 INDEPENDENT VARIABLES

Water depth is an important predictor since it affects light availability at the seafloor as well as sediment accumulation, wave energy and other factors that determine seabed substrate type. The distribution of kelp is therefore restricted to a limited depth interval where there is enough light, where stable hard substrates are available and where the wave energy is not too strong. A bathymetry dataset from EMODnet with 250 m spatial resolution was used, which is the best freely available homogeneous bathymetry dataset for this area.

Wave exposure is one of the most important variables influencing the composition of shallow macroalgae communities. A homogenous wave exposure dataset based on the Simplified Wave Model (SWM) with a spatial resolution of 25 m was used.¹⁶ SWM has been shown to provide an ecologically relevant representation of wave exposure patterns in complex coastal areas.¹⁷¹⁸¹⁹

¹⁶ Isæus, M., "Factors structuring *Fucus* communities at open and complex coastlines in the Baltic Sea."

¹⁷ Eriksson, B. K., et al., "Effects of boating activities on aquatic vegetation in the Stockholm archipelago, Baltic Sea."

¹⁸ Bekkby, T., et al., "Spatial probability modelling of eelgrass (*Zostera marina*) distribution on the west coast of Norway."

¹⁹ Sandman, A., et al., "Spatial predictions of Baltic phyto-benthic communities: Measuring robustness of Generalized Additive Models based on transect data."

Euphotic zone depth is a variable which strongly limits the vertical distribution of all benthic macroalgae species, including kelp. Derived euphotic zone depth (described in Section 3.1 and 4.1) with 300 m spatial resolution for the years 2016-2024 were used (averaged during summer season when euphotic depth is lower and biologic activity is higher).

Changes in **sea water temperature** may affect the distribution of macroalgae communities.²⁰²¹ For this variable, Sentinel-3 SLSTR sea surface temperature (0.02° spatial resolution) was accessed via the CMEMS data store and averaged over the summer season between 2016 and 2024.

3.2.2 DEPENDENT VARIABLE

A training dataset from SHARK with presence/absence and abundance data for the dependent variables from surveys performed between 2016 and 2024 was used. Since the data were collected during several different surveys with various methods, they were unevenly distributed in the modelling area. Additionally, many observations were made as part of transect surveys, but all points along each transect were only given a single coordinate. The dataset did include unique transect identifiers and length but not compass direction, making it impossible to extrapolate the location of multiple points along the same transect. To account for these data limitations, the observational data was rasterized using a 150 m grid (the longest transect in the dataset was 75 m), and each cell of the resulting raster was attributed with “presence” or “absence” of kelp species. A 150 x 150 m pixel with a single observation of a kelp species was thus attributed with “presence.” All pixels with no observational data were excluded from the dataset. The remaining dataset included 1584 unique observations (with 13% prevalence of the species).

3.2.3 MODEL TRAINING

During model training, the training dataset was segmented 80/20 with stratification, reserving 20% of the dataset for validation. After model comparison and hyperparameter tuning, the final model was trained on 100 % of the kelp presence/absence dataset.

Six different classifiers (table 2) were run in 60 iterations where the random forest classifier resulted in the highest F1 scores in the split validation based on the withheld validation data.²²²³ The random forest classifier was therefore selected for the predictive model in which 100 % of the training dataset was used.

Table 2: Classifiers evaluated in the training of the model and average accuracy.

Modelling method	Average accuracy	Average F1 score
Random forest Classifier	0.95	0.78
Gradient boosting Classifier	0.94	0.74
Extra trees Classifier	0.95	0.78
XG boosting Classifier	0.94	0.76
Adaptive boosting Classifier	0.92	0.66
Logistic Regression Classifier	0.90	0.56

²⁰ Arriaga, O., et al., “The thermal journey of macroalgae: Four decades of temperature-induced changes in the southeastern Bay of Biscay”

²¹ Chefaoui, R. M., et al., “Temporal variability of sea surface temperature affects marine macrophytes range retractions as well as gradual warming.”

²² Breiman, L. “Random forests.”

²³ Cutler, R., et al., “Random forests for classification in ecology”

3.3 WATER QUALITY INDICATORS

The Swedish environmental quality standards for water are regulations that establish the quality that water bodies should have or maintain based on reference values determined for each water body's condition and human impact.²⁴ The reference value represents the condition of the water body in the absence of significant anthropogenic influence – essentially, a baseline or “natural” state. These standards aim to ensure good water quality and ecosystem health, and classification of water bodies are conducted for ecological and chemical status. A water body's ecological and chemical status are reclassified at least every six years into one of the following categories:

- High
- Good
- Moderate
- Unsatisfactory
- Bad

A central concept in this classification is the Ecological Quality Ratio - referred to in Swedish as *Ekologisk Kvot* (EK) - calculated as:

$$EK = \frac{\text{Measured Value}}{\text{Reference Value}}$$

This ratio compares the observed value of a water quality parameter to its reference value. An EK value close to 1 indicates conditions similar to the reference state (typically corresponding to “High” or “Good” classification), while lower values suggest increasing ecological degradation.

This project investigated the feasibility of using remotely sensed data to monitor three variables – chlorophyll a concentration, hypoxic conditions at the seafloor, and Secchi depth - that are important in the classification of Swedish coastal water bodies. These data products can be used for multiple purposes in the future, including classification of water bodies with time intervals, environmental monitoring and anomaly detection.

The analysis was conducted within the framework of coastal water bodies – administrative zones along Sweden's west coast defined by the Swedish Water Authorities.²⁵ These zones are known in Swedish as *kustvattenförekomster* (hereafter referred to as KVF), and are delineated based on ecological characteristics and are used as the primary units for water quality assessment and management.

3.3.1 CHLOROPHYLL A

Chlorophyll a (Chl-a) is a key indicator of phytoplankton biomass and is widely used as a proxy for assessing the trophic status and ecological health of aquatic ecosystems. In the context of Swedish environmental quality standards, Chl-a concentrations are one of the primary biological quality elements used to classify the ecological status of coastal water bodies.²⁶ Elevated levels of Chl-a can indicate eutrophication, often driven by nutrient enrichment from land-based sources, and are associated with reduced water clarity. Importantly, Chl-a is also a primary indicator of algal blooms, including harmful algal blooms (HABs), which can lead to oxygen depletion, toxin production, and negative impacts on fisheries, tourism, and public health. Monitoring

²⁴ Havs och Vatten myndigheten, “Havs- och vattenmyndighetens författningssamling”

²⁵ Vattenmyndigheterna, “Vattenförvaltning.”

²⁶ Havs- och vattenmyndigheten, “Havs- och vattenmyndighetens återrapportering av Miljöövervakningsanslaget.”

Chl-a levels at high temporal and spatial resolution is therefore essential for early detection and management of bloom events.²⁷

In this pilot project, Chl-a concentrations were derived from the CMEMS dataset OCEANCOLOUR_BAL_BGC_L3_MY_009_133, which provides daily Level 3 multi-year observations of ocean colour variables for the Baltic Sea. To retrieve Chl-a from satellite reflectance data, the dataset employs a regionally optimized ensemble of multilayer perception neural net (MLP), known as Chl-a_{ENS3}.²⁸ This algorithm dynamically combines outputs of multiple MLP models trained on different spectral band combinations, using a novelty index to weight each model's contribution based on the similarity of the input spectrum to the training data. The final Chl-a estimate is computed as weighted average:

$$Chl - a_{ENS3} = \frac{\sum_{i=1}^{N=3} w_i \cdot Chl - a_{MLP}}{\sum_{i=1}^{N=3} w_i}$$

Where, Chl-a_{ENS} is the final Chl-a concentration (mg/m³) by the ensemble, N is the number of MLP models used in the ensemble, Chl-a_{MLP} is the Chl-a value predicted by the i -th MLP model (each trained on a different combination of spectral bands), w_i is the weight assigned to each model's output, calculated as, $w_i = 1/\eta_i$ where η_i is the novelty index for model i .

In this formulation, models that are more familiar with the input data (i.e., have lower novelty index) are given greater influence in the final results. This adaptive weighting approach improves retrieval accuracy in optically complex waters like the Baltic Sea, where standard algorithms often struggle due to high concentration of Coloured Dissolved Organic Matter (CDOM) and variable atmospheric conditions.

The Chl-a data were analysed to assess spatial and temporal variability within the project's area of interest, with a focus on their potential to support classification of water bodies according to the Swedish Water Agency's coastal water zones. These zones provide reference conditions and thresholds for Chl-a concentrations, enabling classification into one of five ecological status categories as mentioned above.

3.3.2 HYPOXIC CONDITIONS

Hypoxia at the seafloor is a parameter used in the assessment of ecological status under the Swedish national environmental quality standards. Hypoxia is defined as oxygen concentrations ≤ 2 ml/l sea water and anoxic conditions as ≤ 0 ml/l sea water.²⁹ Hypoxia at seafloor is quantified as the proportion of seafloor with hypoxia. In the national Swedish monitoring program, this parameter is measured at the deepest areas of monitored coastal water zones in profiles from the surface to the seafloor. However, only a few of the Swedish coastal water zones are included in the monitoring.

Hypoxic and anoxic conditions at the seafloor were modelled with the Copernicus Marine Service Baltic Monitoring and Forecasting Centre (BAL MFC). Bottom oxygen is one of the parameters in the Copernicus product "Baltic Sea Analysis Forecast BGC 003 007" which contains the operational forecasts for the biogeochemical conditions in the Baltic Sea including Skagerrak (fig. 1). The product has a spatial resolution of 2 x 2 km and is updated at a frequency of two times per day.

²⁷ Brando, V. E., et al., "Phytoplankton bloom dynamics in the baltic sea using a consistently reprocessed time series of multi-sensor reflectance and novel chlorophyll-a retrievals."

²⁸ Ibid.

²⁹ Cato, I., et al., "A new approach to state the areas of oxygen deficits in the Baltic Sea."

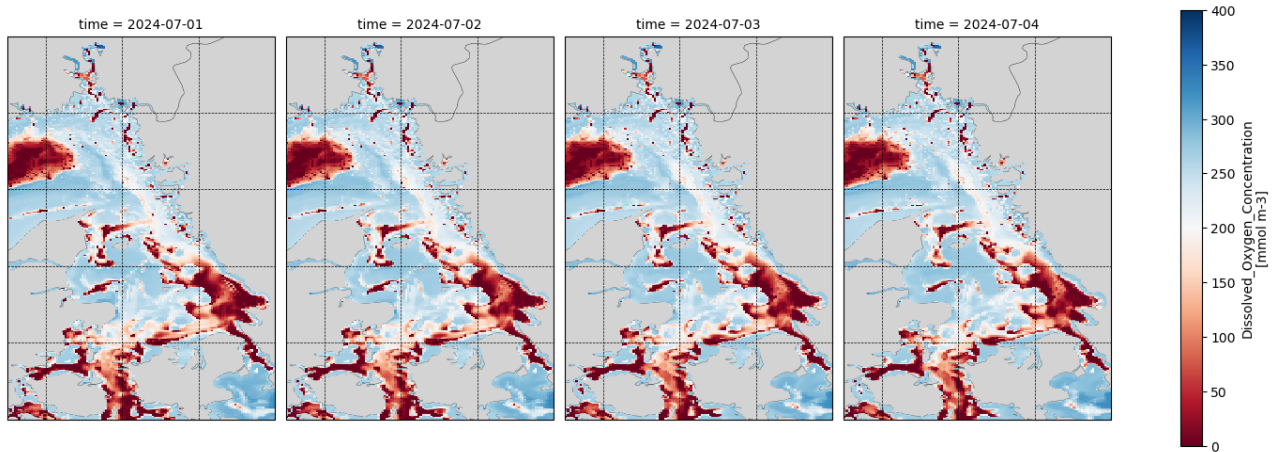


Figure 1. Dissolved oxygen concentration at seafloor (mmol/m³) during four days in July 2024 from Copernicus Marine Service Baltic Monitoring and Forecasting Centre (BAL MFC).

Hypoxic (and anoxic) conditions at seafloor were calculated from the daily bottom oxygen parameter averaged over the full time series (July 2024). Oxygen at seafloor was converted from mmol O₂/m³ into ml O₂/l sea water. The proportion of seafloor with hypoxic conditions within the coastal water zones were delineated as areas with ≤ 2 ml O₂/l sea water.

3.3.3 SECCHI DEPTH

Secchi depth (SD) is a traditional and widely used measure of water transparency, representing the depth at which a Secchi disk - a circular, black-and-white disk - is no longer visible from the surface. It serves as a proxy for water clarity and is influenced by the presence of phytoplankton, suspended sediments, and dissolved organic matter.³⁰ In Swedish coastal waters, Secchi depth is one of the parameters used to assess ecological status under national environmental quality standards.³¹

In remote sensing applications, Secchi depth can be estimated using satellite-derived reflectance data, offering a scalable and efficient alternative to *in situ* measurements. In this project, Secchi depth was derived using a band ratio algorithm based on remote sensing reflectance (Rrs) values from the CMEMS dataset OCEANCOLOUR_BAL_BGC_L3_NRT_009_131.

The applied algorithm was specifically tailored for the Baltic Sea and estimates water transparency, expressed in the unit of metres (m), using a blue-to-red reflectance ratio.³² In optically clear waters, a greater proportion of blue light is reflected compared to red light. As water becomes more turbid - due to increased concentrations of organic and inorganic matter - this ratio decreases, reflecting the enhanced absorption and scattering of shorter wavelengths.

The algorithm is based on the ratio of remote sensing reflectance at 490 nm (blue) to 709 nm (red), and is expressed as:

$$SD = \left(\frac{Rrs(490)}{Rrs(709)} \right)^{0.697} \cdot 2.137$$

³⁰ Preisendorfer, R., "Preisendorfer, Rudolph W., "Secchi disk science: Visual optics of natural waters 1."

³¹ Havs- och vattenmyndigheten, "Havs- och vattenmyndighetens författningssamling."

³² Alikas, K. & Kratzer, S., "Improved retrieval of Secchi depth for optically-complex waters using remote sensing data"

This empirical relationship was applied using the variables RRS490 and RRS708_75 of the CMEMS dataset (see Section 2.2). The resulting Secchi depth estimates provide a reliable and scalable method for monitoring water clarity across the study area and can be integrated into broader environmental assessment and marine planning frameworks.

4 RESULTS

This section shows the results of each data product presented in Section 3. Each Jupyter Notebook included in the code repository investigates a specific spatial and temporal data subset, and the results presented here reflect that. However, users of the notebooks can easily edit the spatial and temporal subsets, making the code applicable to the entire Baltic Sea drainage basin.

4.1 EUPHOTIC ZONE DEPTH AND EUPHOTIC ZONE DELINEATION

To demonstrate the methodology, a subset of satellite data from May to September 2024 covering the study area was processed. Initial visualisation of daily $K_d(490)^{-1}$ revealed data gaps due to cloud cover (fig. 2), which were addressed using a 7-day forward filling technique to improve spatial continuity across the time series (fig. 3).

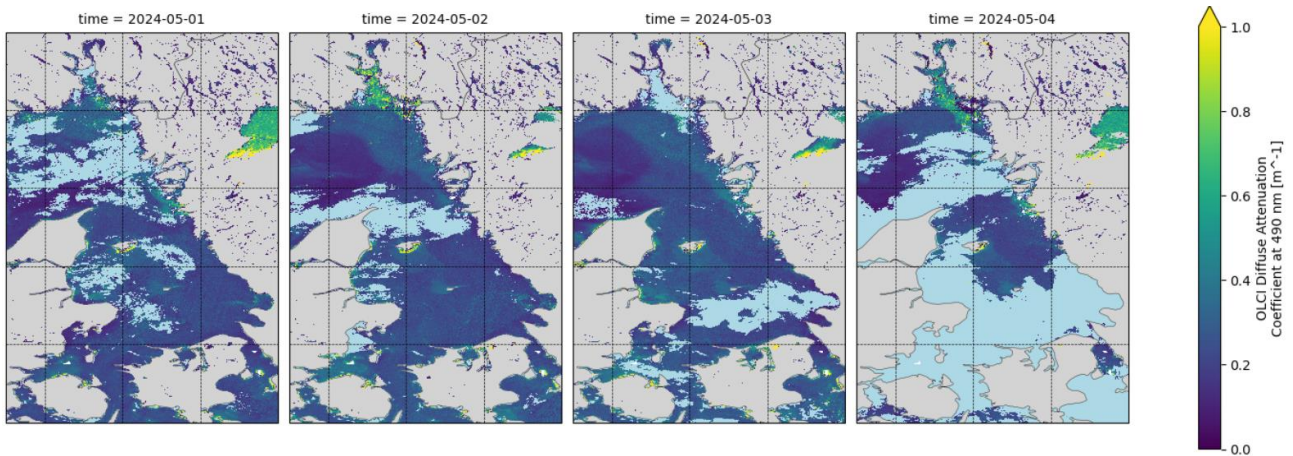


Figure 2. The figure shows the first four days of the original $K_d(490)^{-1}$ data loaded from CMEMS. [Link to high resolution image.](#)

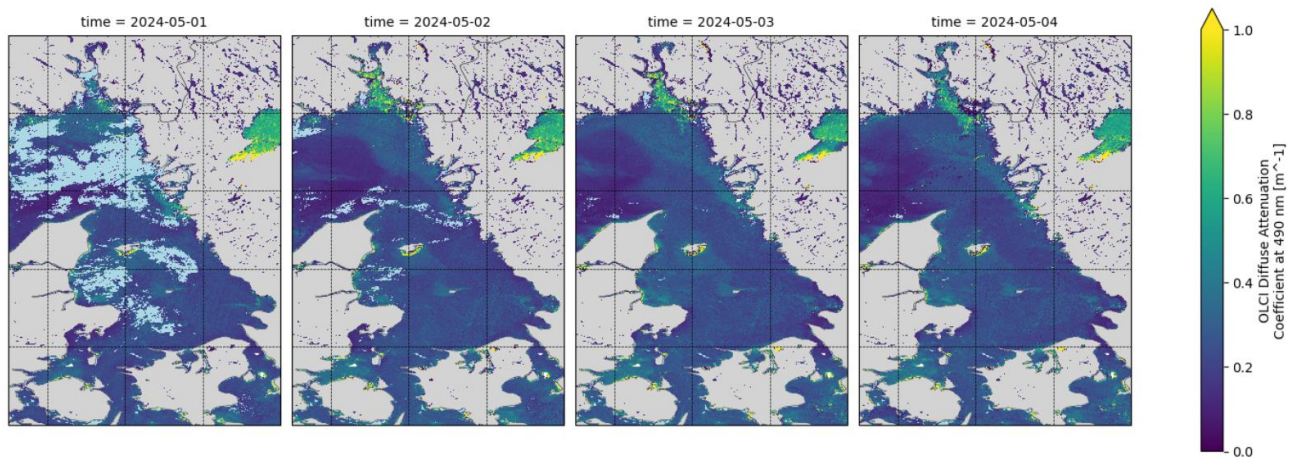


Figure 3. The figure shows the first four days of the same data subset after the application of the forward filling algorithm. [Link to high resolution image.](#)

As described above (see section 3.1), $K_d(490)^{-1}$ was converted into more interpretable units – meters of light penetration in the water column – and the results were stored in a new variable `Zeu_depth`, the euphotic zone depth. The resulting values were visualised as maps, revealing spatial variability in light penetration, with shallower euphotic depths in turbid coastal areas and deeper penetration in clearer offshore waters (fig. 4).

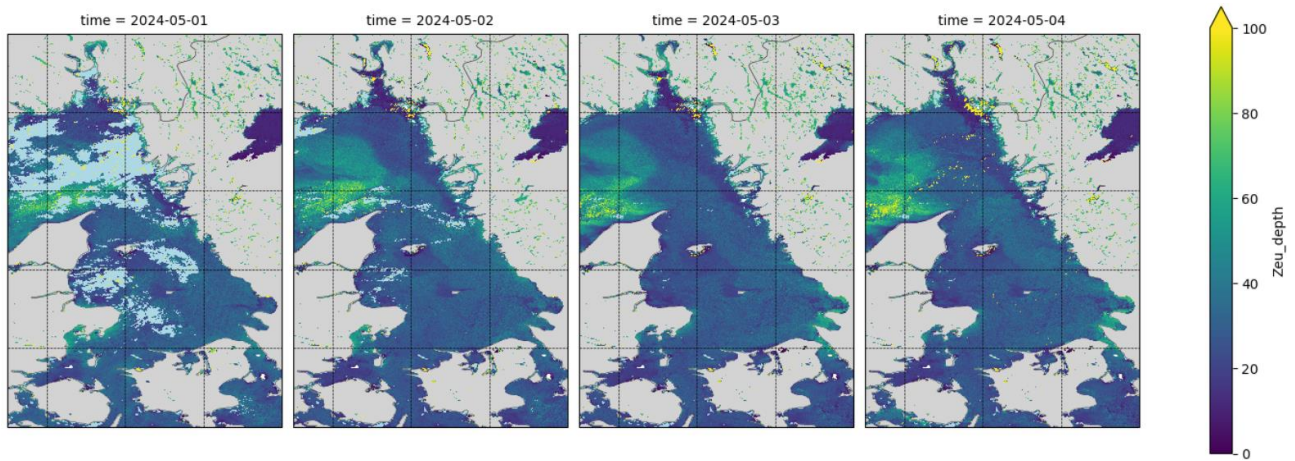


Figure 4. The first four days of the data subset after the application of Kratzer et al. 2003 algorithm for the derivation of the euphotic zone depth. Data is visualised on a scale of 0-100 m penetration in the water column. [Link to high resolution image.](#)

These initial results provided a foundation for further analysis, including the delineation of the euphotic zone at the seafloor (`Zeu_zone`) (fig. 5) and the calculation of the percentage of light availability at the seafloor (`Zeu_percent`).

Euphotic Zone

May 02 2024

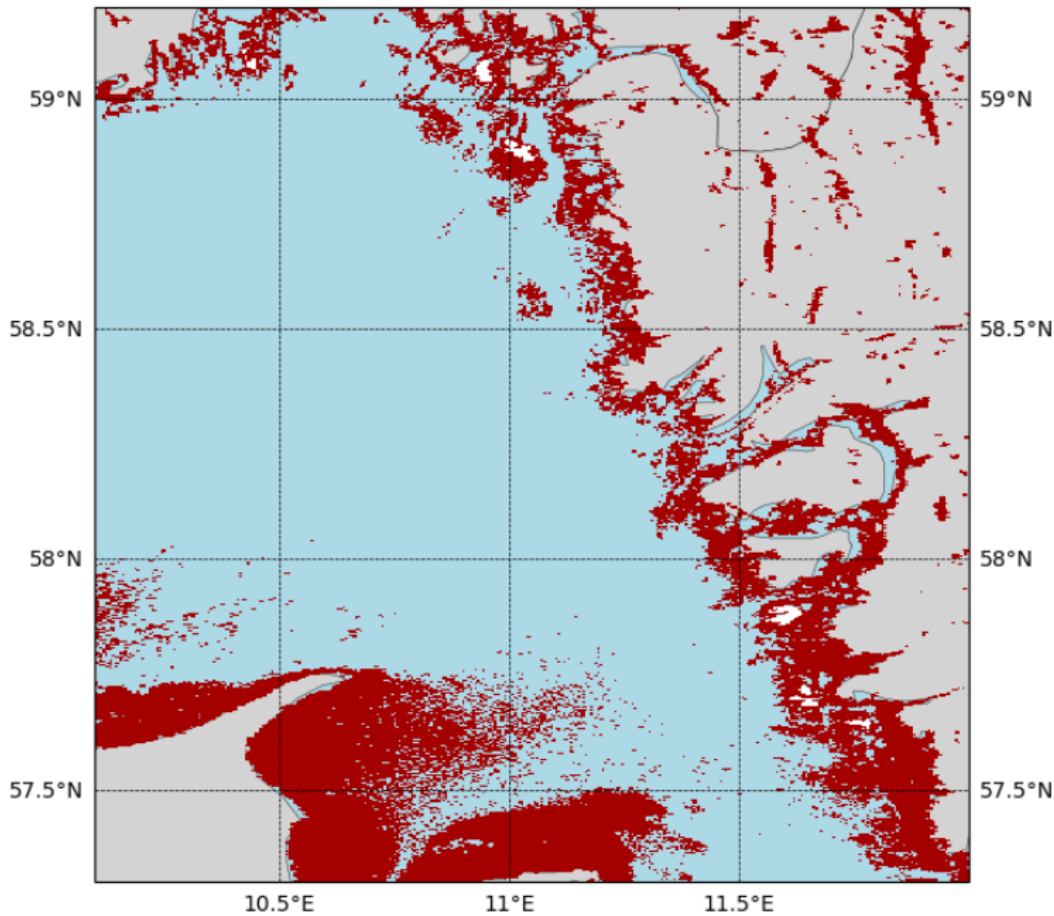


Figure 5. A time slice of a single day in the data series showing the binary variable `Zeu_zone` in the study area. Red pixels symbolize areas where at least 1% of light incident on the water surface reached the seafloor on this day. [Link to high resolution image.](#)

To illustrate temporal patterns, the dataset was resampled to weekly and seasonal averages (fig. 6). Spatial gradients in light availability were observed, with higher `Zeu_percent` values in shallower coastal regions and lower values in deeper and presumably more turbid areas.

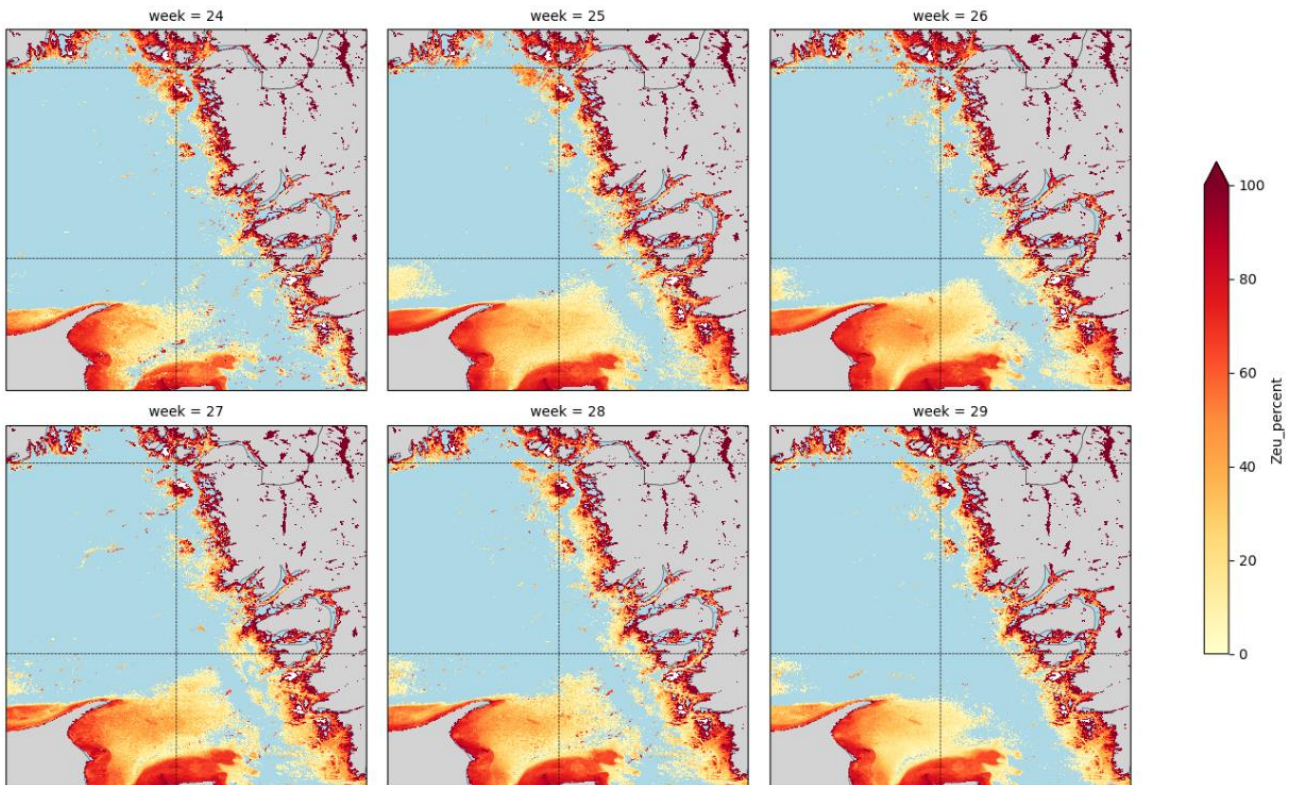


Figure 6. The map series shows the weekly average of the percentage of light incident on the seafloor in the study area, with darker red area denoting pixels with a higher percentage of light reaches the seafloor. [Link to high resolution image.](#)

Additionally, the notebook allows for the user to generate an interactive map of the daily data product using xcube-viewer – including Zeu_depth, Zeu_zone, and Zeu_percent – for the entire time series selected. This interactive map allows the user to zoom in to see local detail, select a specific pixel or polygon to investigate how the underwater light environment changed during the time series, and change the background map for geographic context.

The results presented serve as an example of how the workflow can be applied to other time periods and regions using the same methodology.

4.2 MACROALGAE HABITAT DISTRIBUTION

To illustrate how remote sensing data can be utilized alongside other continuous environmental variables to predict the distribution of key marine vegetation communities, the data product kelp habitat distribution (as an example of macroalgae habitat distribution) is presented in this section. This data product may be used in predictions of available of kelp habitat, the distribution of which will change over time with changes in environmental variables such as euphotic zone depth as a result of climate change or a change in nutrition load.

The macroalgae habitat distribution modelling, based on the random forest classification model had a performance with an average accuracy of 95% and f1 score of 0.78, based on split validation with 20% withheld validation data.

The feature importance of independent variables indicated that SWM was the most significant predictor (the predictor that explained most of the variation in the dependent variable), followed by (in descending order of importance) water depth, euphotic zone depth and Sentinel-3 SLSTR Sea Surface Temperature (fig. 7).

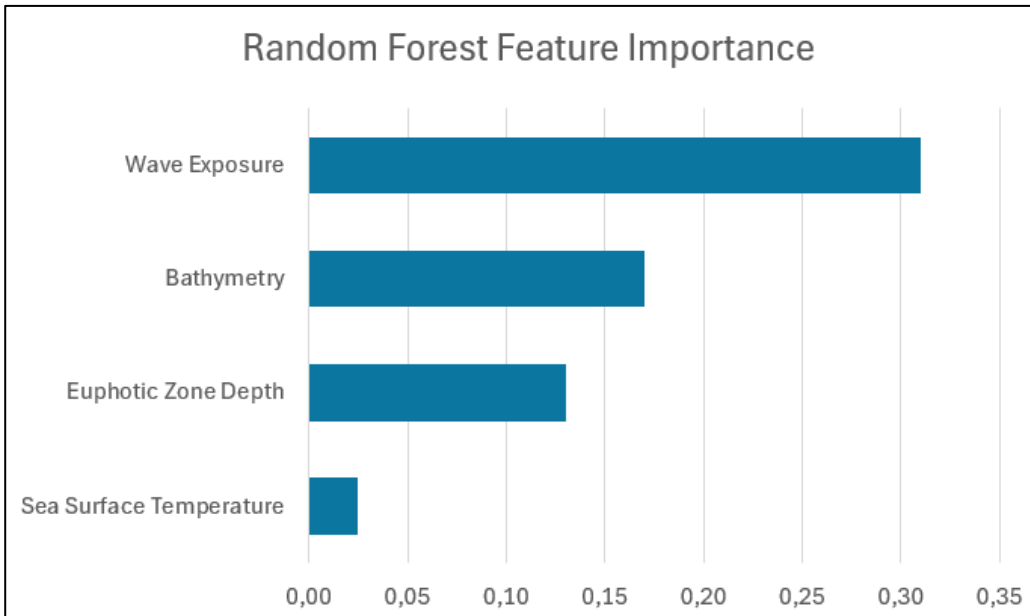


Figure 7. The relative feature importance of the independent variables in the random forest classification model.

The prediction based on the entire training dataset was performed at ca 150 m spatial resolution. It indicates that kelp habitats are primarily concentrated along the coast with the largest habitat concentration in zones in the outer coastal areas (fig. 8) and around islands in the outer archipelago such as the Koster islands and Väderöarna where hard substrates are more abundant and euphotic zone depth is greater than in inshore areas (fig. 9).

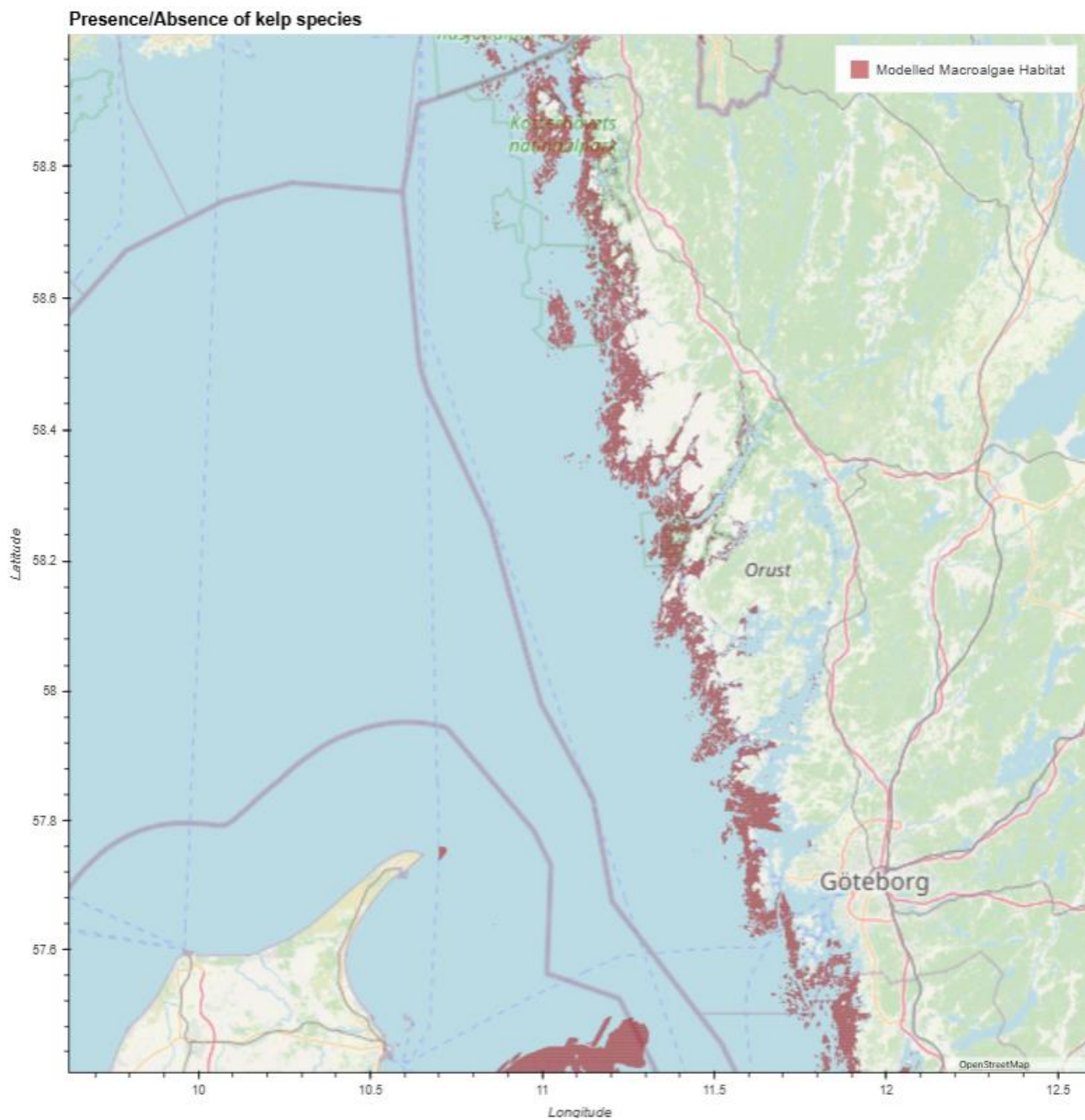


Figure 8. Prediction of kelp habitat in ca 150 m resolution. The predicted presence of kelp habitat within a raster cell means that kelp habitat may occur somewhere in the cell. When generated in the Jupyter notebook, this map is interactive and allows for panning and zooming. [Link to high resolution image.](#)

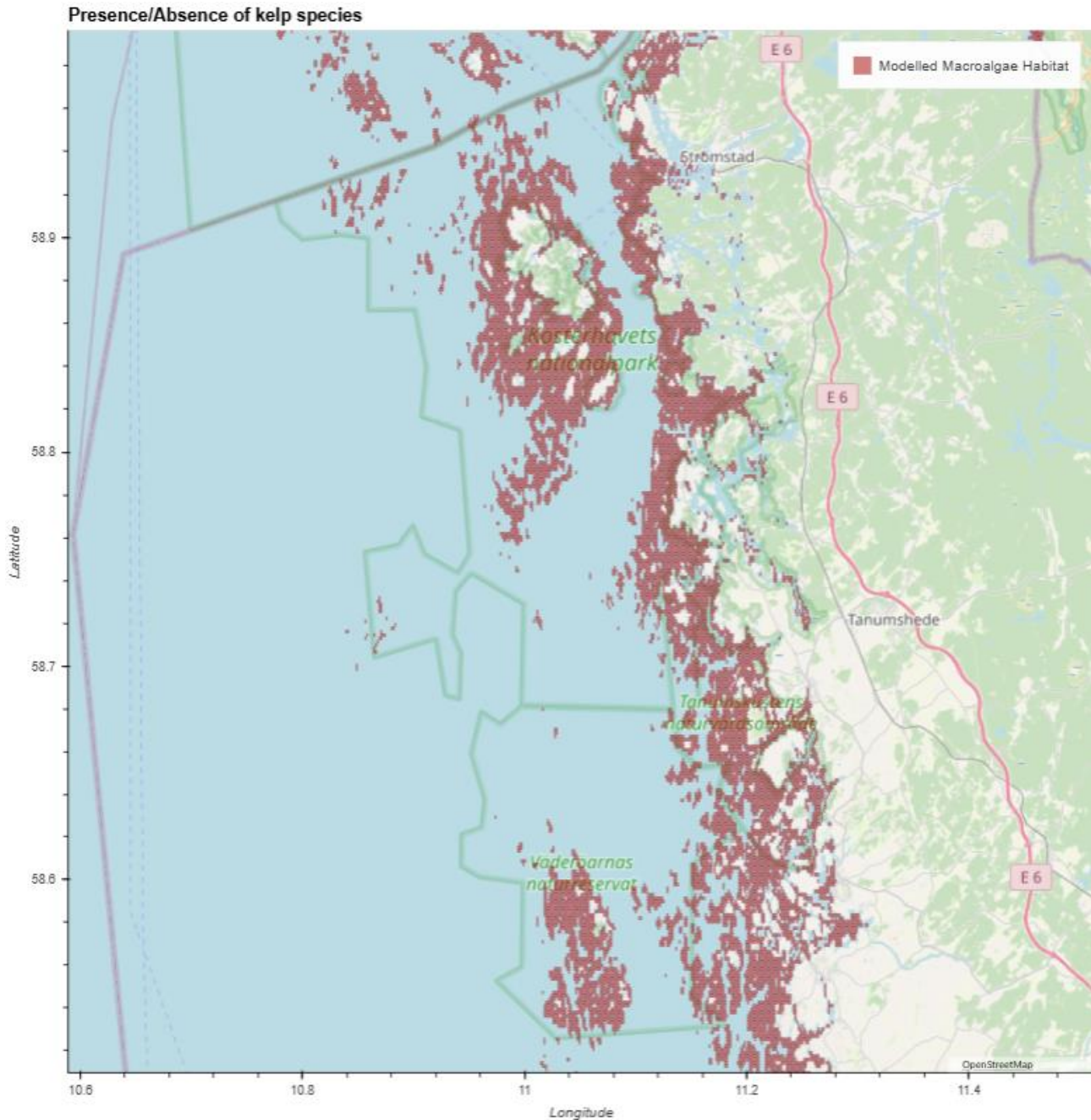


Figure 9. A zoomed image of the kelp habitat prediction showing the modeled habitat zones in the outer archipelago around the Koster islands and Väderöarna where hard substrates are prevalent, and the euphotic zone is more widespread. [Link to high resolution image.](#)

The kelp habitat prediction illustrates the predicted distribution of kelp habitat during the period 2016-2024. A continuously updated model will illustrate changes over time in the distribution available habitat for kelp. Since kelp are perennial, long lived species, their distribution is more likely to change due to gradual changes in euphotic zone depth over time than due to occasional short deviations. Sea surface temperature was the least important predictor of the distribution of kelp habitat on this scale, but the importance of the variable may increase with a rise in water temperature and/or at larger geographic areas.

4.3 WATER QUALITY INDICATORS

To illustrate the applied approach, the following sections describe the derivation and classification of the three different water quality indicators - Chlorophyll a, hypoxic conditions, and Secchi depth – across the study area, in alignment with the Swedish environmental quality standards. Together, these indicators provide complementary insights into the ecological status of coastal waters, supporting a more integrated and robust assessment of environmental conditions while also demonstrating the potential of the Digital Twin Ocean approach as a scalable tool to support data-driven marine planning and environmental monitoring.

4.3.1 CHLOROPHYLL A

A case study was conducted for the study area during July 2024, using the variable “Mass concentration of chlorophyll a in sea water” from the CMEMS product. Data were extracted for the period 1-31 July 2024, resulting in a 31-layer daily data cube representing spatial grids of Chl-a concentration.

Due to frequent cloud cover, many pixels were missing leading to data gaps. To address this, a forward-filling technique was applied, allowing valid observations to be propagated forward in time for up to seven days. This approach improved spatial completeness while limiting the use of outdated data. A temporal mean was then calculated across the 31-day period to produce a composite map of average summer Chl-a concentrations.

To illustrate the impact of the forward-filling method, figures 10 and 11 show Chl-a concentrations for the first four days of July 2024, before and after the gap filling, respectively. Note that, while only these four days were visualised for demonstration purposes, the entire monthly dataset was used in the subsequent classification analysis.

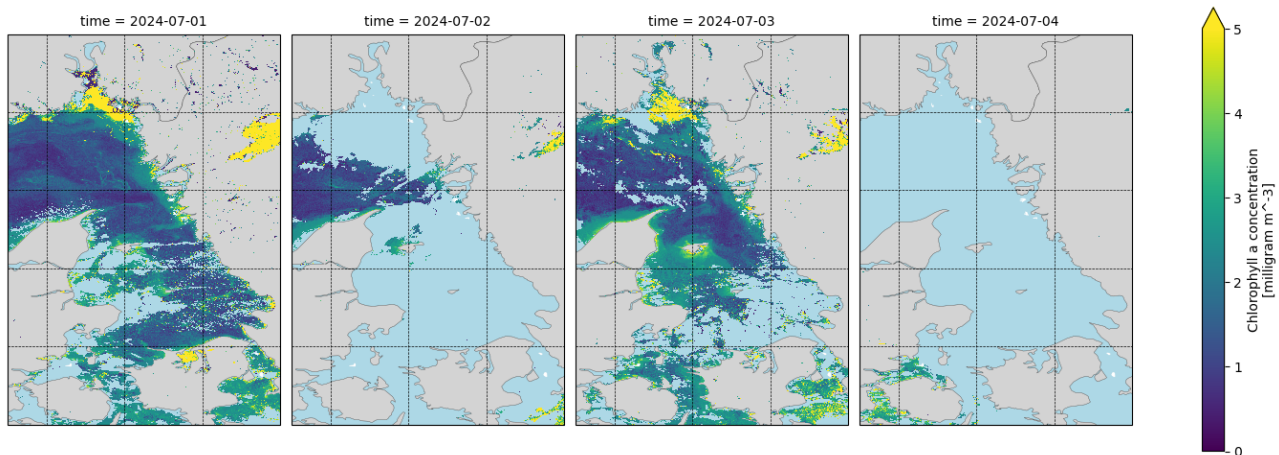


Figure 10. The figure shows the first four days of the original Chl-a concentration data loaded from CMEMS. [Link to high resolution image.](#)

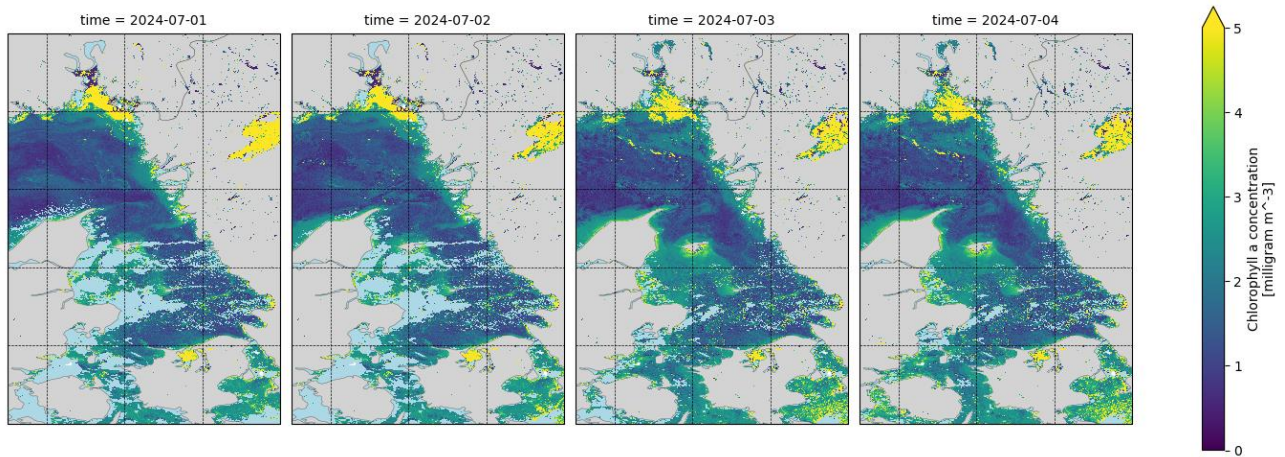


Figure 11. The figure shows the first four days of the same data subset as figure 10 after the application of the forward filling algorithm. [Link to high resolution image.](#)

The mean Chl-a dataset was spatially clipped to the boundaries of coastal water bodies (KVF) using shapefiles provided by the Swedish Water Agency, providing nine distinct coastal water types within the study area. Each water body was processed individually to enable localized analysis (fig. 12).

Chlorophyll a (Chl-a) in sea water KVF:5

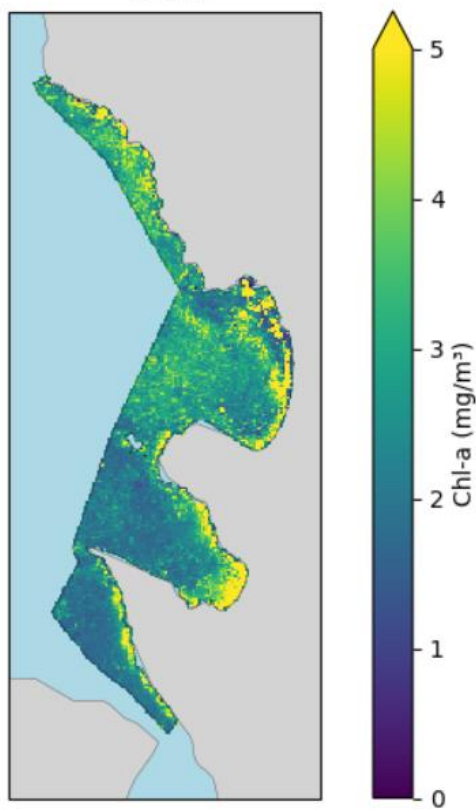


Figure 12. The figure shows the mean Chl-a concentration (mg/m^3) for July 2024 in the coastal water zone KVF 5, which was selected as an example for illustration purposes. [Link to high resolution image.](#)

To assess ecological status, the mean Chl-a concentrations were classified according to the HVMFS 2019:25 guidelines. For each water body type, a reference value (Rv) and a set of threshold ratios (High, Good, Moderate, Unsatisfactory, Bad) were used to calculate the Ecological Quality Ratio (EK). Each pixel was assigned to one of five ecological status classes based on its EK value. The classification was applied to each water body, and the results were visualised as individual maps (fig. 13). For each area, a summary table was generated showing the number and percentage of pixels in each class.

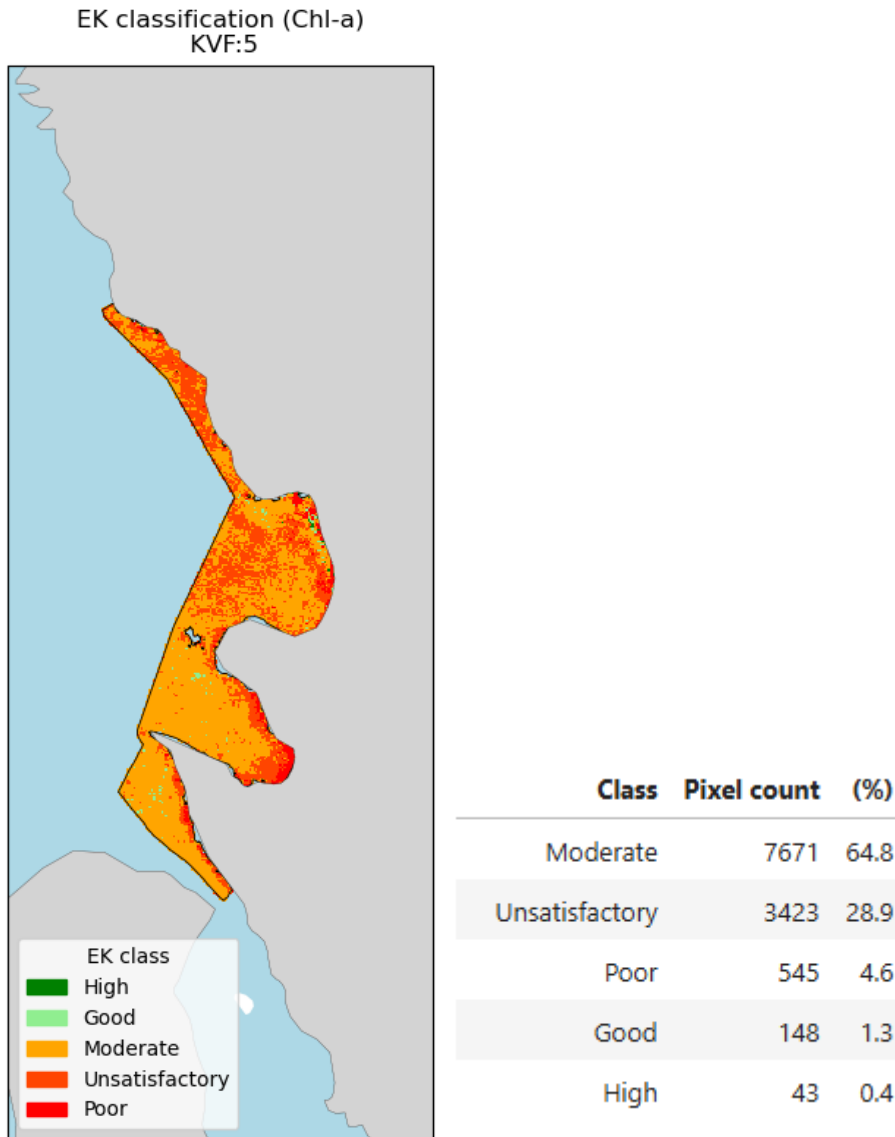


Figure 13. The figure shows the ecological status classification based on the ecological quality ratio (EK) of the mean Chl-a concentration (mg/m³) for July 2024 in the coastal water zone KVF 5 (left), along with a summary table showing the percentage of pixel in each class (right). [Link to high resolution image.](#)

The classification results revealed clear spatial patterns, with elevated Chl-a concentrations – and potential signs of eutrophication – particularly evident in bays and nearshore areas.

A combined classification map was produced to provide an overview of ecological status across all coastal water bodies in the study area (fig. 14). This map supports the identification of eutrophication hotspots and can inform regional water management strategies.

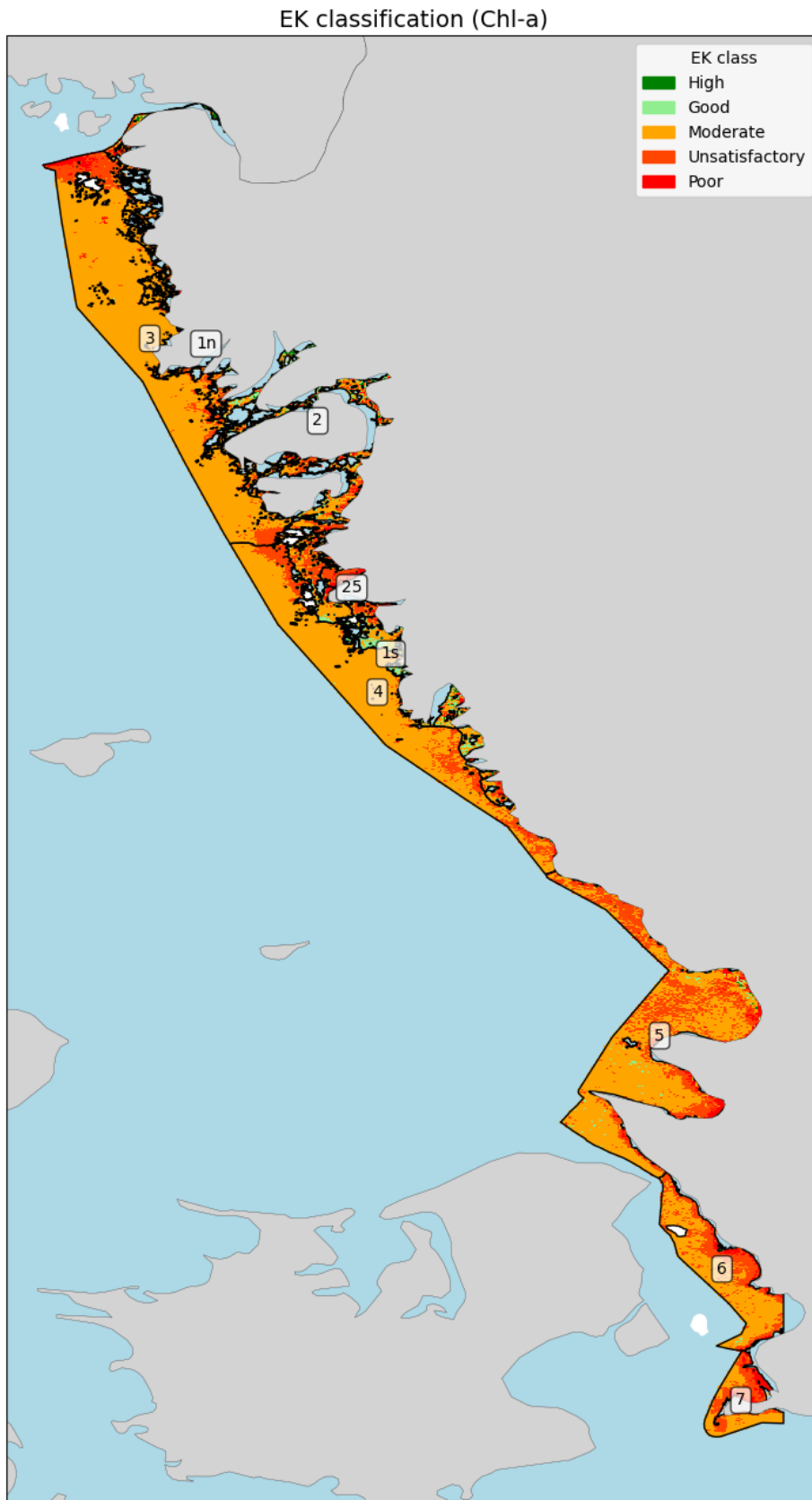


Figure 14. The figure shows the ecological status classification based on the ecological quality ratio (EK) of the mean Chl-a concentration (mg/m^3) for July 2024 in the coastal water zone within the study area. [Link to high resolution image](#).

4.3.2 HYPOXIC CONDITIONS

A case study showing the potential to monitor the proportion of seafloor with hypoxic or anoxic conditions using remotely sensed data was conducted for the month of July 2024. The input data used to create this product was the variable “Sea floor dissolved oxygen concentration” from the Baltic Sea Biogeochemistry Analysis and Forecast product in CMEMS – a numerically modeled data product with 2 x 2 km spatial resolution produced by the biogeochemical model ERGOM one way coupled to a Baltic Sea set up of the NEMO ocean model (see Section 2.2 for references to the data source).

The data was initially averaged over the selected temporal subset and translated from its original units of mmol/m^3 to units of ml/l . The classification of hypoxic and anoxic seafloor conditions was conducted first on the entire study area, resulting in no defined anoxic areas, but significant areas of hypoxic areas (fig. 15).

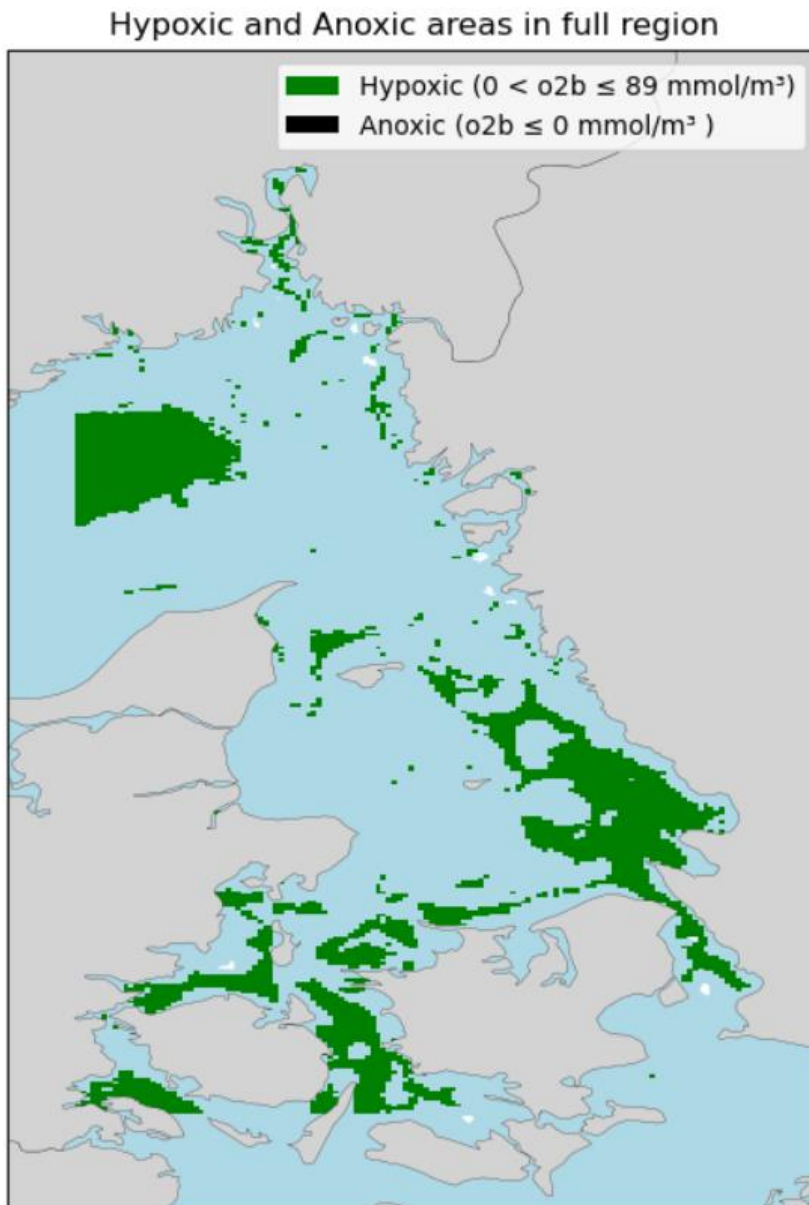


Figure 15. The entire study area with hypoxic and anoxic pixels classified for a temporal average of July 2024. [Link to high resolution image.](#)

The resulting classified dataset was further clipped to the boundaries of coastal water bodies (KVF), enabling localized analysis (Fel! Hittar inte referensälla.16).

KVF 3
Hypoxic: 10.4%, Anoxic: 0.0%

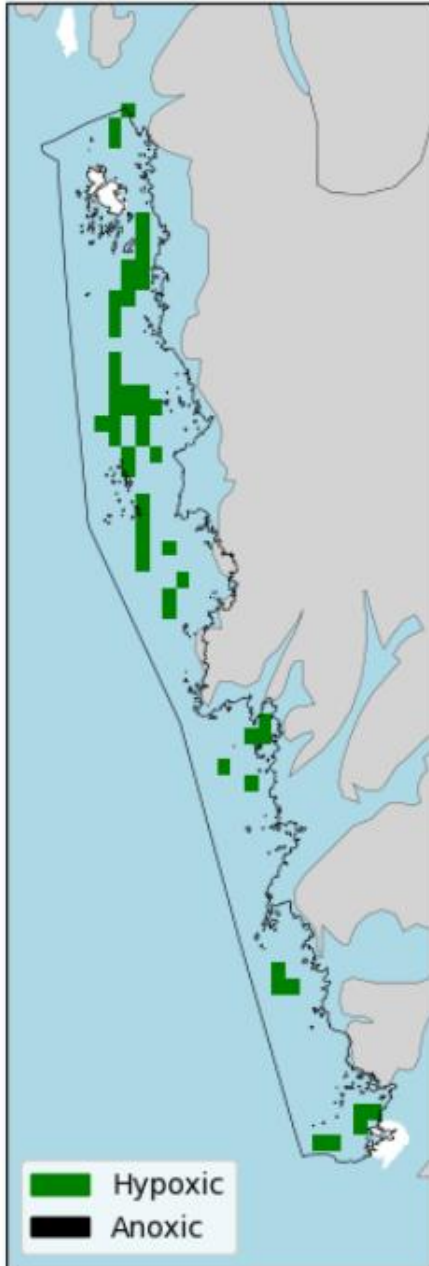


Figure 16. Classified pixels with hypoxic and anoxic conditions in KVF 3, as well as percentage of total KVF area classified as each.
[Link to high resolution image.](#)

There were no areas identified with anoxic conditions (≤ 0 ml O_2 /l sea water) in any of the included coastal water zones from during the period, but areas with hypoxia (≤ 2 ml O_2 /l sea water) occurred in some coastal water zones, mainly in southern Kattegat where hypoxia was widespread in the coastal water zones 5 and 6 (table 3).

Table 3. Proportion of seafloor with hypoxic and anoxic conditions.

Coastal water zone	Total area (km ²)	Hypoxic area (km ²)	Anoxic Area (km ²)	Poportion of seafloor with hypoxic conditions (%)	Poportion of seafloor with anoxic conditions (%)
1n	700	24	0	3.4	0.0
1s	604	8	0	1.3	0.0
2	504	20	0	4.0	0.0
25	104	0	0	0.0	0.0
3	2048	212	0	10.4	0.0
4	1436	80	0	5.6	0.0
5	1616	756	0	46.8	0.0
6	640	332	0	51.9	0.0
7	272	0	0	0.0	0.0

This classification can be used to provide a continuously updated estimate of the proportion of seafloor with hypoxia and anoxia within the coastal water zones and the product provides a good overview of the entire sea basins as well. Smaller inshore areas with local hypoxia or anoxia will however most likely be missed due to the spatial resolution. In the future, this data product can be produced at both the daily scale as well as varying temporal averages (such as the euphotic zone data product), to investigate change over time in relation to other environmental conditions.

4.3.3 SECCHI DEPTH

To estimate Secchi depth (SD) across the study area, remote sensing reflectance (R_{rs}) data at 490 nm and 708.75 nm were accessed from the CMEMS dataset by utilizing the RRS490 and RRS708_75 variables (see Section 2.2). The analysis focused on summer period from July 1 to July 31, 2024, aligning with the time frame used for the Chl-a data product. Similarly to the Chl-a analysis, the dataset included 31 daily observations.

Initial visualisations of the reflectance data revealed significant data gaps due to cloud cover. To address this, a forward-filling technique was applied along the time dimension, allowing missing values to be filled using the most recent valid observation, with a maximum gap of 7 days.

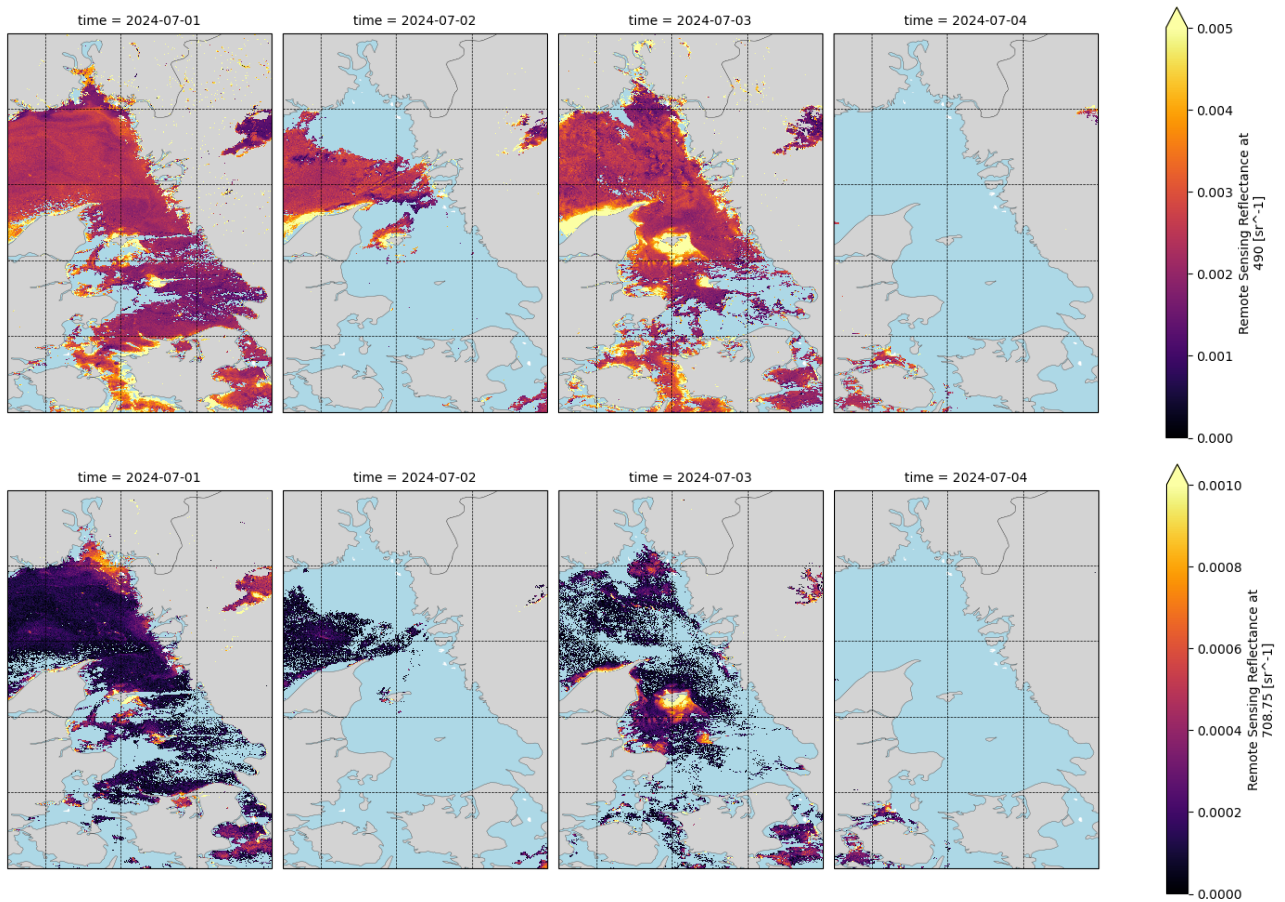


Figure 17. The figure shows the first four days of the original reflectance data loaded from CMEMS at 490 nm (top) and at 708.75nm (bottom). Link to high resolution images: [Reflectance at 490nm](#), [Reflectance at 708.75nm](#).

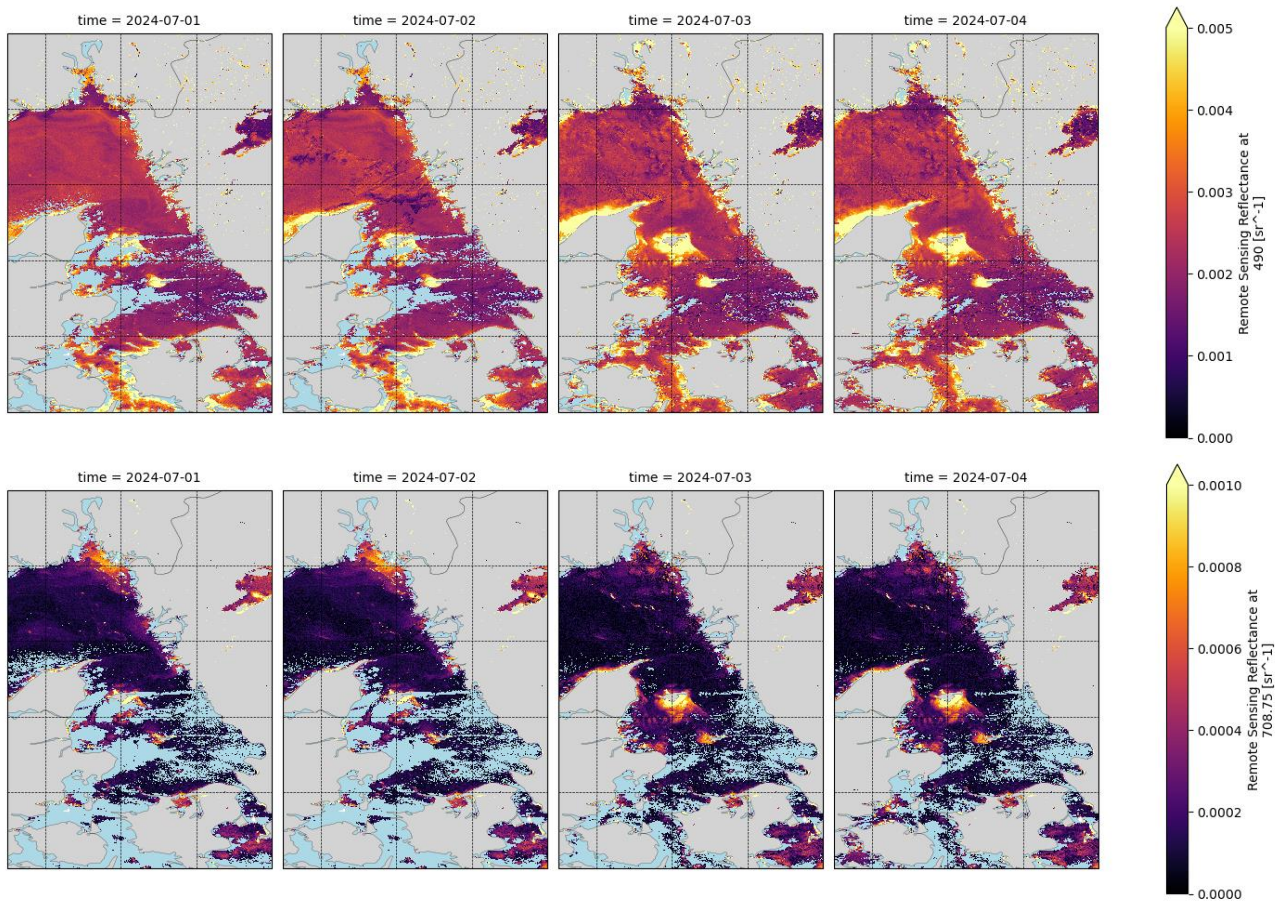


Figure 18. The figure shows the first four days of the same data subset of reflectance at 490 nm (top) and at 708.75 nm (bottom) as figure 17 after the application of the forward filling algorithm. Link to high resolution images: [Reflectance at 490nm \(forward filled\)](#), [Reflectance at 708.75nm \(forward filled\)](#).

This significantly improved the continuity of the dataset and using the forward-filled reflectance values, Secchi depth was calculated. To ensure realistic values, a maximum cap of 20 m was applied to the results, and for visualisation purposes a subset of the first four days of July was selected (fig. 19).

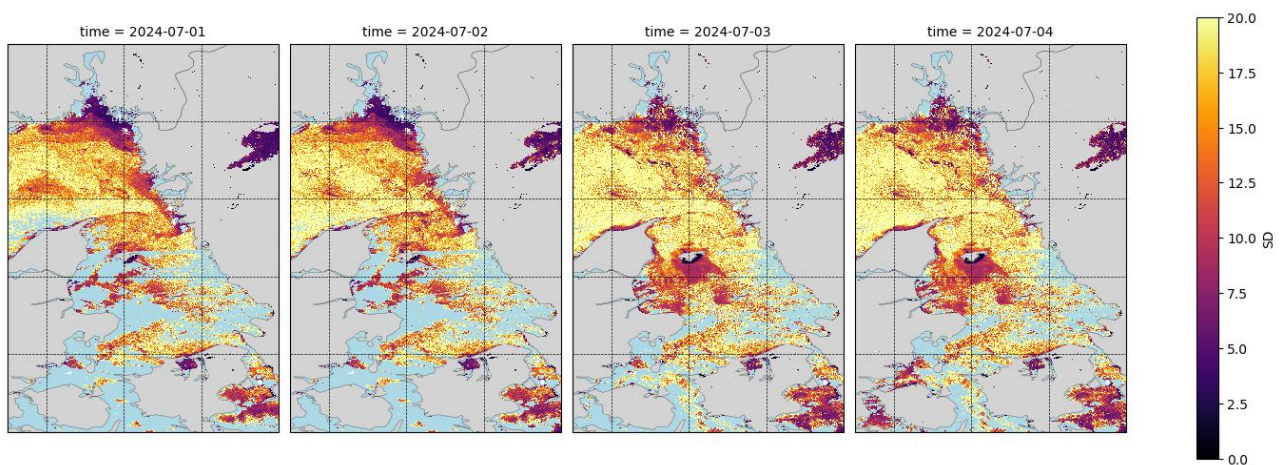


Figure 19. The figure shows the estimated Secchi depth (m) for the first four days, based on the remote sensing reflectance ratio at 490 nm and 708.75 nm. [Link to high resolution image.](#)

The results indicated a wide range of water clarity conditions across the study area. Higher transparency values were generally observed in offshore regions, while lower values concentrated near the coast, particularly in areas influenced by riverine input or biological activity.

Although the applied algorithm may overestimate Secchi depth – particularly in clearer pelagic waters where values frequently reached the upper cap of 20 m - the spatial patterns of water transparency remained consistent with ecological expectations. High transparency was found in offshore areas where Chl-a concentrations were low, and reduced clarity was observed nearshore where Chl-a was elevated. This suggests that the algorithm performs reliably under the conditions observed in July 2024, though further validation across different bloom scenarios and environmental conditions is recommended.

To support regional water quality assessments, the mean Secchi depth dataset was spatially clipped to the Swedish coastal water bodies (KVF) using official shapefiles (fig. 20).

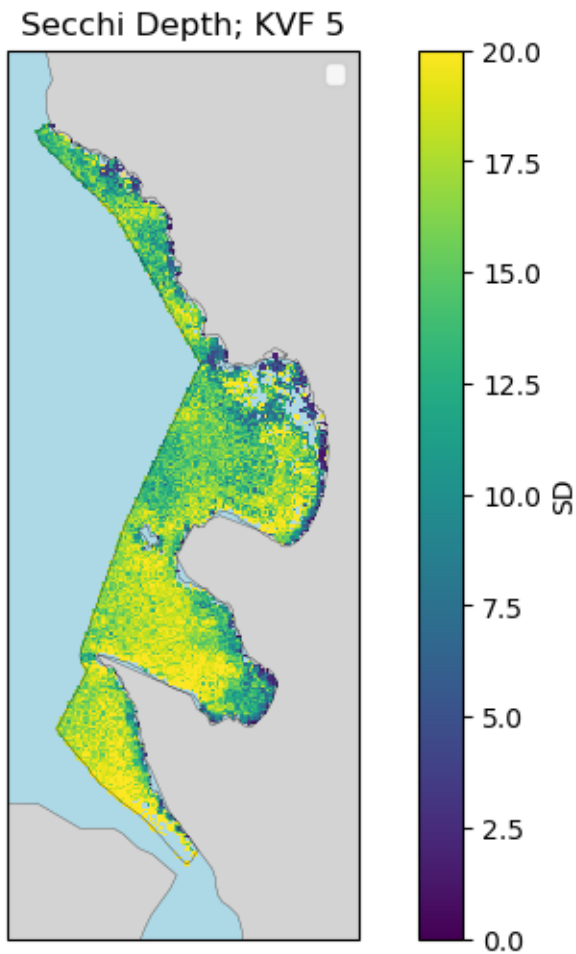


Figure 20. The figure shows the mean Secchi depth (m) for July 2024 in the coastal water zone KVF 5, which was selected as an example for illustration purposes. [Link to high resolution image.](#)

The calculated Secchi depth values were then classified according to the HVMFS 2019:25 guidelines. This allowed for targeted analysis within ecologically defined zones and supports integration with national environmental monitoring frameworks. The results were visualised as a series of maps, one for each coastal water body (fig. 21). For each area, a summary table was generated showing the pixel count and percentage distribution across the five classes for each water body. This provides a quantitative basis for evaluating water clarity and identifying areas that may require further investigation or management intervention.

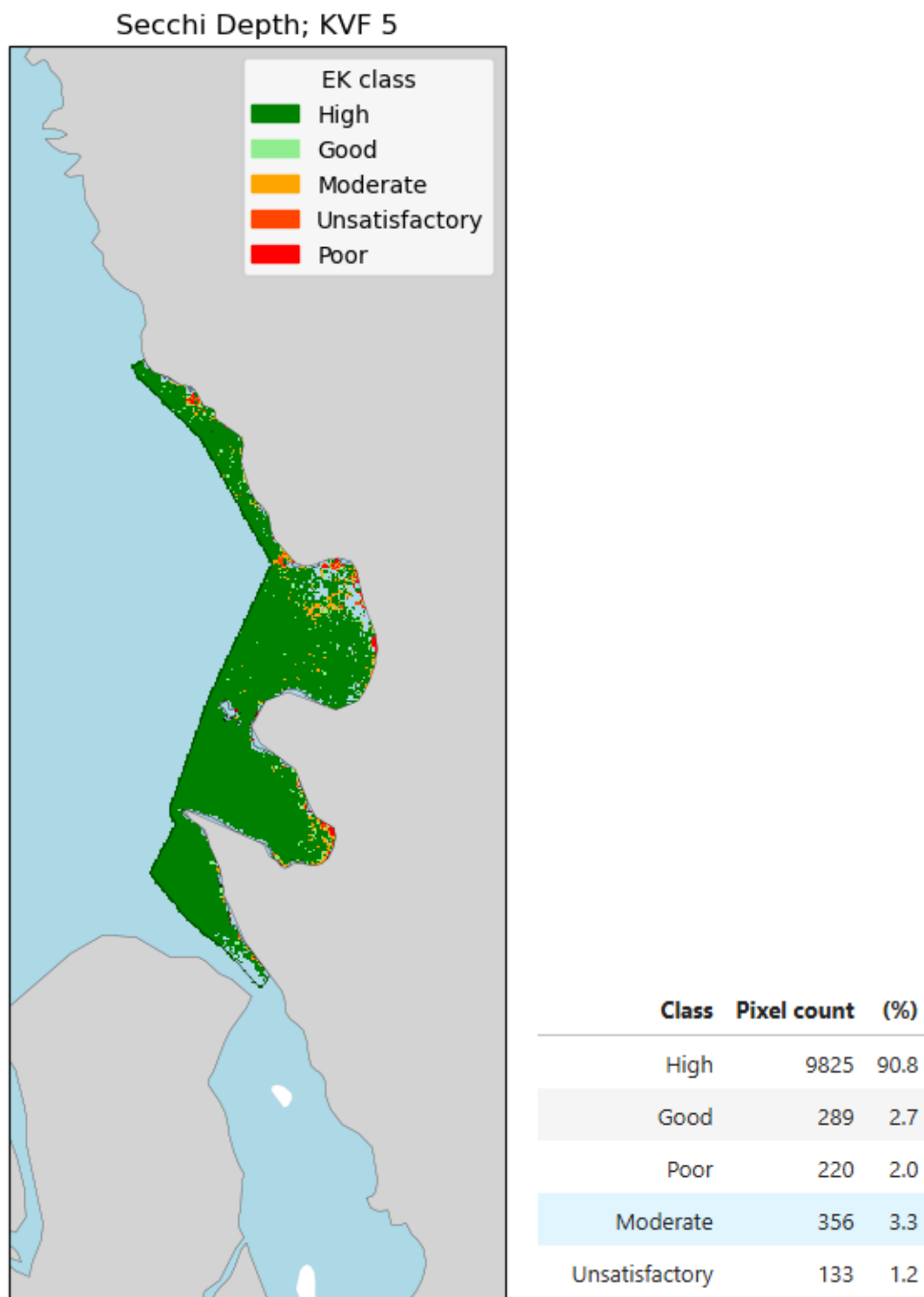


Figure 21. The figure shows the ecological status classification based on the ecological quality ratio (EK) of the mean Secchi depth (m) for July 2024 in the coastal water zone KVF 5 (left), along with a summary table showing the percentage of pixel in each class (right). [Link to high resolution image.](#)

It is worth noting that as the algorithm tends to overestimate Secchi depth, high ecological classifications observed across some areas should be interpreted with caution. These results may not fully reflect actual conditions and should be considered indicative rather than definitive.

To provide a comprehensive overview of water clarity conditions, the ecological classification results for all coastal areas within the study area were compiled into a single map (fig. 22). This visualisation highlights regional patterns of water transparency at a broader spatial scale, enabling stakeholders to identify areas of concern, prioritize monitoring or mitigation efforts, and promote collaboration among different administrative bodies.

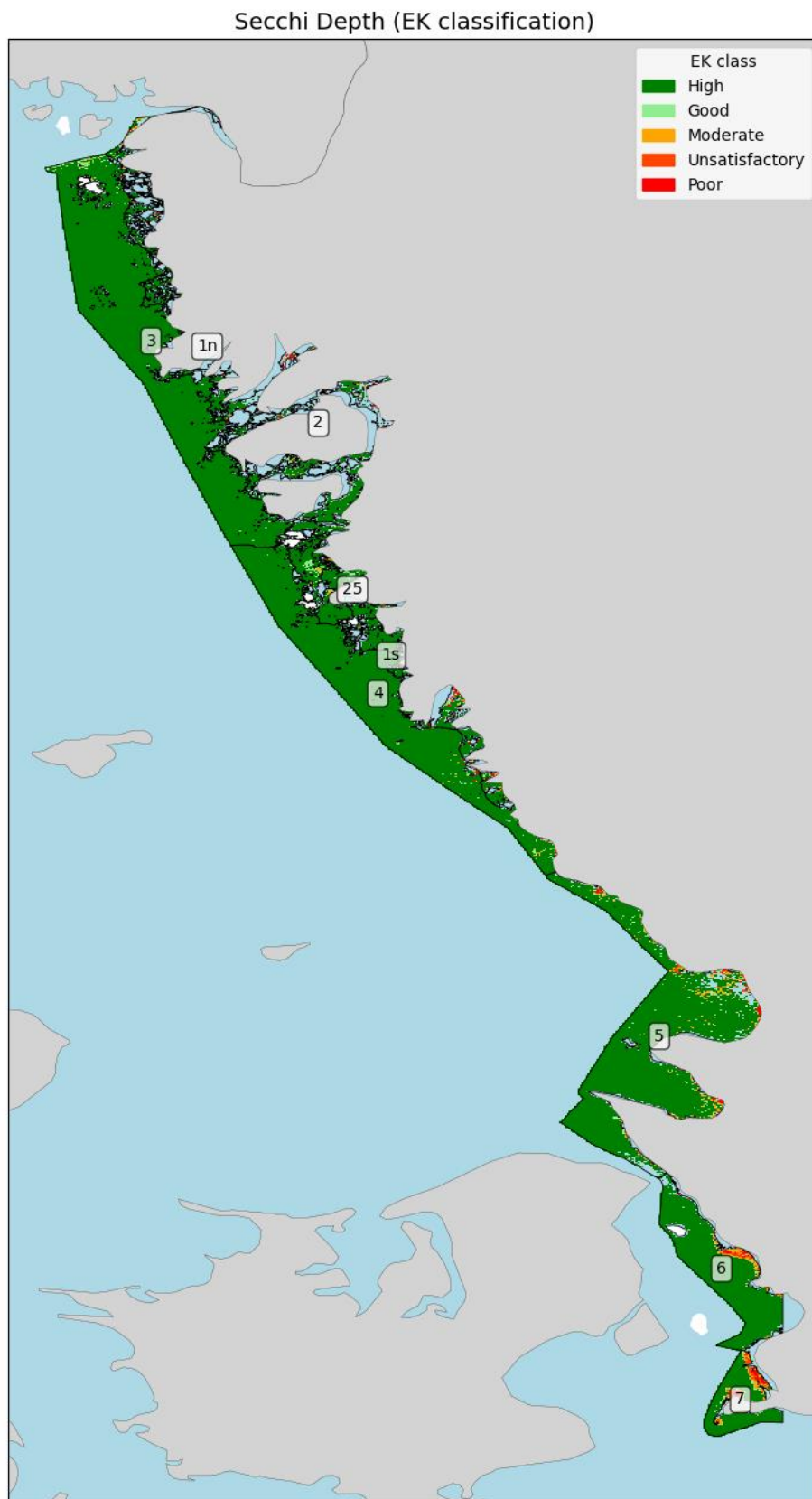


Figure 22. The figure shows the ecological status classification based on the ecological quality ratio (EK) of the mean Secchi depth (m) for July 2024 in the coastal water zone within the study area. [Link to high resolution image.](#)

5 DISCUSSION

The primary goal of this pilot project was to show the potential for deriving actionable insights from satellite- and remotely sensed data. During preliminary discussions with stakeholders, it was clear that this could best be achieved by adding vertical (phenomena occurring under the water surface) and temporal (up-to-date data as well as longer time-series) dimensionality to the general understanding of the marine environment.

STRENGTHS AND LIMITATIONS OF THE APPROACH

A major strength of this pilot lies in its accessibility and replicability. By leveraging open-source Python libraries and the WEKEO JupyterLabs platform, the project shows that advanced marine data products can be developed without requiring high-performance local computing infrastructure. This significantly lowers the barrier for municipalities and regional agencies to explore and adopt Digital Twin Ocean (DTO) methodologies. Furthermore, the use of interactive tools like xcube-viewer enhances transparency and user engagement. Stakeholders can explore the data themselves, promoting both easy access and data literacy among non-experts. In addition, the modular structure of the Jupyter Notebooks allows for easy adaptation to other regions, time periods, or environmental variables. This makes the approach highly scalable and suitable for broader national or transboundary marine planning efforts.

Despite these advantages, limitations were encountered. Satellite-derived datasets are inherently affected by cloud cover, particularly in northern latitudes, which led to temporal gaps in the data. While forward-filling techniques were employed to improve continuity, these methods inevitably introduced a degree of uncertainty. Additionally, some of the remote sensing algorithms used in this project - particularly those for estimating euphotic zone depth and Secchi depth - were originally developed for Sweden's east coast. Given the distinct optical and ecological conditions of the Skagerrak and Kattegat regions, the performance of these algorithms may vary, and their outputs should be interpreted with caution. Perhaps most significantly, the absence of *in situ* validation data for most products limited the ability to assess their absolute accuracy. The macroalgae habitat model stands out as an exception, benefiting from a robust observational dataset that allowed for more confident validation.

IMPLICATIONS FOR MARINE PLANNING, MONITORING AND POLICY

The results of this pilot underscore the value of a DTO approach in supporting evidence-based decision-making. In the context of marine spatial planning, the delineation of euphotic zones and macroalgae habitats provides critical information for identifying ecologically valuable areas and assessing the potential impacts of coastal development. For environmental monitoring, the ability to generate both near-real-time and historical data products enables continuous tracking of key indicators such as chlorophyll a and Secchi depth. This capability is particularly important for the early detection of eutrophication events and harmful algal blooms, which can have significant ecological and socio-economic consequences. In this regard, DTOs also hold promise as early warning systems for environmental risks such as hypoxia or sediment plumes, enabling more proactive and timely management responses - an especially valuable function in climate-sensitive coastal zones.

Beyond direct applications in planning and monitoring, the DTO framework also supports broader policy and governance goals. By aligning with existing frameworks such as the Swedish Water Agency's coastal water body classifications, the approach enhances the potential for integration into national monitoring programs and supports adaptive management strategies. Moreover, the standardized and open-access nature of the tools and data encourages collaboration across municipalities, national agencies, and research institutions, fostering more coherent and harmonized marine governance. Finally, the visual and interactive nature of DTO outputs offers a powerful means of engaging the public and communicating complex marine processes

in accessible ways. This can be particularly useful in public consultations or educational contexts, helping to build broader support for marine conservation and sustainable planning initiatives.

6 FURTHER DEVELOPMENT

There is much room for development in future efforts to both improve the data products presented in this project as well as work towards a full deployment of a maritime planning digital twin. To make further results as useful as possible for the end users, all further development should be considered in consultation with local, regional, and national partners who work closely with maritime planning and/or the marine environment. The project team considers the following to be the most important focus areas in future work:

DEPLOYMENT OF A FULL-SCALE DIGITAL TWIN OCEAN FOR MARITIME PLANNING

This project, along with other similar efforts, has shown the potential for actionable insights to be derived from satellite- and remotely sensed data in maritime planning. While the code repository provided allows for non-expert users to their own spatial and temporal subsets, there remains a technical barrier to entry which could in turn lead to non-distributed solutions that advance or stagnate at varying paces. A full deployment of a DTO for Maritime Planning could provide near real-time datasets that are easily downloadable or streamable via mainstream geospatial platforms, allowing for simple user interaction and greater reach. Additionally, such a platform would be centrally managed and maintained, and allow for more timely development. Such a platform would require significant computing resources as well as a dedicated maintenance team.

SCIENTIFIC DATA PRODUCT IMPROVEMENT

The methods and algorithms used to derive actionable datasets can and should be updated at regional levels as the science progresses to a better understanding of the connection between remotely sensed data variables and real-world phenomenon. For example, an algorithm used to derive the euphotic zone depth from $K_d(490)^{-1}$ and validated in the Gulf of Bothnia based on the optical sensing depth may not be directly applicable to the Skagerrak. However, scientific advances may soon provide a better data-driven method to derive the optical sensing depth from remotely sensed data, allowing future work to automatically tailor the algorithm used to derive euphotic zone depth to a specific region. These advances should be included in a future deployment of any digital twin of the underwater environment.

FURTHER PRODUCT DEVELOPMENT

The data products created in this pilot were necessarily limited in scope due to time constraints and the relevant expertise of the project group and thus focus primarily on environmental factors. While an improved understanding of the changing environmental status in coastal and offshore waters is important and can be greatly expanded in the future, future efforts may also investigate the following areas:

- 1) **Direct human impacts on coastal and offshore waters.** Products in this category could include:
 - Noise pollution models based on real-time shipping data, offshore wind farms, and harbor installations. This would be particularly useful for maritime planning if combined with modelled environmentally sensitive zones.
 - The impact of offshore wind farms on coastal physical parameters. The impact of large-scale offshore wind farms is not well understood at this time, and insights into how these installations may impact physical parameters closer to shore would be greatly beneficial to coastal communities.

- The current DTO prototype focuses primarily on environmental variables. For comprehensive marine spatial planning, future iterations could integrate socioeconomic data (e.g., fisheries, tourism, shipping lanes) to support multi-use planning.
- 2) **Extreme event modelling.** Products in this category could include:
- Machine/deep learning models trained on historical data to predict the impact of extreme weather events on coastal communities and environmental parameters. Models could also be created to predict the impact of more gradual events such as sea level rise and rising water temperatures.
 - Early detection models. Models could be built to focus on specific remote sensing variables or derived environmental parameters to detect aberrant behavior
- 3) **Expansion of water quality indicators to include nutrients.**
Incorporating nutrient monitoring - particularly nitrogen and phosphorus - in the DTO framework would significantly enhance the utility of the water quality product. These nutrients are central to the assessment of ecological status and are key drivers of eutrophication as well as cyanobacterial and diatom bloom dynamics. Including them would provide a more holistic view of water quality conditions, enabling better prediction and management of bloom events. It would also strengthen the link between observed biological responses (e.g., Chl-a levels, Secchi depth) and their underlying causes, improving both the interpretability and actionability of the data product.
- 4) **Identification of suitable areas for macroalgae cultivation using habitat modeling.**
Building on the macroalgae habitat distribution model, future work could adapt the existing machine learning framework to identify optimal sites for macroalgae cultivation. Macroalgae, such as kelp, are not only valuable as a sustainable biomass resource but also play a role in nutrient uptake and water quality improvement. By incorporating additional variables into the model developed in this pilot, such as nutrient concentrations, proximity to nutrient sources (e.g., river mouths or wastewater outlets), and potential anthropogenic pressures, the model could be refined to highlight areas where cultivation would be both ecologically and environmentally beneficial.

INTERACTIVE MARITIME PLANNING TOOLS

The development of a web-based interactive toolbox would make a DTO for maritime planning more actionable and allow non-expert users to interact with and better understand the data products. This toolbox would need to be built on a fully deployed DTO platform and could include tools for users to visualise datasets together, pull in their own data, and draw further insights from their geospatial orientations. Such a toolbox could also allow “on demand” modelling scenarios – allowing user to select a temporal and spatial subset of multiple datasets of interest and then run a machine learning model (such as the model presented in this project for predicting macro-algae habitats) on their area of interest. Placing the correct tools in the hands of decision makers and the people actively working with these questions would likely lead to greater insights and faster development.

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APPENDIX A: GITHUB REPOSITORY

All code used to produce the data products described in this technical report are included along with external data sources in the linked GitHub repository. All code is presented as Jupyter notebook tutorials that walk the user through the subsetting of data and creation of data products. The notebooks can be imported into a WEkEO workspace and run by any user with a free user account. The code is licensed under an MIT license and is free to use for further development.

[GitHub Repository](#)

URL: <https://github.com/WSP-Sweden/DTO-Skagerrak/tree/main>

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